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THE AVERAGED CONTROL SYSTEM OF FAST-OSCILLATING CONTROL SYSTEMS*

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Abstract. For control systems that either have a fast explicit periodic dependence on time and bounded controls or have periodic solutions and small controls, we define an *average control system* that takes into account all possible variations of the control, and prove that its solutions approximate all solutions of the oscillating system as the frequency of the oscillations tends to infinity. The dimension of its velocity set is characterized geometrically. When it is maximum the average system defines a Finsler metric, which is not twice differentiable in general. For minimum time control, this average system allows one to give a rigorous proof that averaging the Hamiltonian given by the maximum principle is a valid approximation.

Key words. averaging, control systems, small control, optimal control, Finsler geometry

AMS subject classifications. 34C29, 34H05, 49J15, 93B11, 93C15, 93C70, 53B40

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1. Introduction.

We consider either a "fast-oscillating control system" (see (1))

$$\dot{x} = u_1 X_1\left(\frac{t}{\varepsilon}, x\right) + \cdots + u_m X_m\left(\frac{t}{\varepsilon}, x\right),\ \|u\| \leq 1,$$

where all X_i's are 2π-periodic with respect to t/ε, or a "Kepler control system" (see (46))

$$\dot{\xi} = f_0(\xi) + v_1 f_1(\xi) + \cdots + v_m f_m(\xi),\ \|v\| \leq \varepsilon,$$

where all solutions of $\dot{\xi} = f_0(\xi)$ are periodic.

Averaging techniques for conservative—periodic or not—ordinary differential equations (ODEs) date back at least to Poincaré; see [2, sect. 52] or [23] for recent expositions. Roughly speaking, on a fixed interval, the solutions of $\dot{x} = F(t/\varepsilon, x)$ differ from those of $\dot{x} = \overline{F}(x)$ by a term of order ε, with $\overline{F}$ the average of F with respect to its first argument.

If u or v above is assigned to be a fixed function of state and time (or computed from additional state variables as in $u = \alpha(p, x),\ \dot{p} = g(p, x)$), then these techniques for ODEs can be applied to give an approximation at first order with respect to small ε of the movement of the slow variables. Averaging is usually used in this way in control theory: in vibrational control [19], fast-oscillating controls are designed and averaging techniques allow analysis and proof of stability; in the same way, averaging solves stability and path planning questions in control of mechanical systems (see, for instance, [8]); in [12, sect. 5], high frequency control is used to approach a nonflat system by a flat one; one may also mention many applications to control [21, 18, 20]

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of the work [17] that mimics Lie brackets by highly oscillatory controls along the original vector fields. A common feature of these references is that the use of oscillations "creates" new independent controls used for design. The application of averaging in optimal control of oscillating systems [10, 13, 14, 7] derives also from ODE techniques but is closer to the framework of this paper because oscillations are present in the system instead of being introduced by the control. Very interesting results are obtained when applying averaging to the Hamiltonian equations arising from the Pontryagin maximum principle. For instance, in [7], the authors have studied in this way the problem of minimal energy transfer between two elliptic orbits; extremals are the same as those giving the geodesics of a Riemannian metric. Again, averaging introduces "new independent controls": Riemannian geodesics are minimizers of a problem where all velocity directions are allowed, whereas the velocity set of the original system at each point had positive codimension. The same averaging computation may be applied to the Hamiltonian differential equation obtained for minimum time, but, since this differential equation is discontinuous, there is no theoretical justification for averaging in that case.

Our contribution is to introduce a different way of averaging that takes into account all possible variations of the control—hence the control strategy can be decided *after* performing averaging—and to prove that it has satisfying regularity properties and is a good first order approximation of the above systems as $\varepsilon \to 0$. This gives, as a side result, a justification of the use of averaging for minimum time in [13, 14]. This procedure also "creates new independent control," i.e., increases the dimension of the velocity set that we characterize in terms of the original vector fields. When this dimension is maximum, the average system defines a Finsler metric [3] on the manifold, whose geodesics are the limits of minimum time trajectories for the original systems as $\varepsilon \to 0$. This Finsler metric is in general not twice differentiable (hence it is indeed not a Finsler metric in the sense of [3]); however, we prove that, at least in the less degenerate case, the Hamiltonian system governing extremals, although it is not locally Lipschitz, generates a flow on the cotangent bundle. Low thrust planar orbit transfer belongs to this less degenerate case.

The average control system may be used for other purposes than optimal control; for instance, [4] designs a Lyapunov function for feedback control in the average system and uses it for the oscillating systems; indeed, the present work was developed out of comparing feedback control based on a priori chosen Lyapunov functions with minimum time control for low thrust orbital transfer.

Our paper is organized as follows. The construction and results are developed for a "fast-oscillating control system" in section 3, transferred in section 4 to "Kepler control systems," and applied to minimum time orbit transfer in the planar 2-body problem in section 5.

2. Notation and conventions.

2.1. M is a smooth connected manifold of dimension n; its tangent and cotangent bundles are denoted by $\mathrm{T}M$ and T^*M. One may assume for simplicity $M = \mathbb{R}^n$, $\mathrm{T}M = \mathbb{R}^n \times \mathbb{R}^n$, $\mathrm{T}^*M = \mathbb{R}^n \times (\mathbb{R}^n)^*$, and, for $x \in M$, $\mathrm{T}_xM = \mathbb{R}^n$, $\mathrm{T}^*_xM = (\mathbb{R}^n)^*$.

For $v \in \mathrm{T}_xM$, $p \in \mathrm{T}^*_xM$ (or any v, p taken in a vector space and its dual), we denote by $\langle p, v\rangle$ (rather than $p(v)$) their duality product.

2.2. If E is a subset of a vector space V, then $E^\perp$ is its annihilator, i.e., the vector subspace of its dual V^* made up of all p's such that $\langle p, v\rangle = 0$ for all v in E.

2.3. We assume that M is endowed with an arbitrary Riemannian distance d. If $M = \mathbb{R}^n$, just choose the canonical Euclidean distance.

In local coordinates, $\|.\|$ and $(.\,|.)$ stand for the canonical Euclidean norm and scalar product. On a compact coordinate chart, $k_1\|x-y\| \le d(x,y) \le k_2\|x-y\|$ for some positive k_1, k_2 (Lipschitz equivalence). We also denote operator norms by $\|.\|$.

2.4. S^1 is $\mathbb{R}/2\pi\mathbb{Z}$. For θ in S^1 (an angle), we denote by $\mu(\theta)$ the unique real number in $[0, 2\pi)$ such that $\mu(\theta) \equiv \theta \mod 2\pi$. For a real number $s \in \mathbb{R}$, we denote the angle it represents by $s \bmod 2\pi$; it belongs to the quotient S^1.

Maps $S^1 \to E$ (arbitrary set) are identified with 2π-periodic maps $\mathbb{R} \to E$. For instance, if f is a map defined on S and $\tau \in \mathbb{R}$, we write $f(\tau)$ instead of $f(\tau \bmod 2\pi)$; the average of f is denoted by $\frac{1}{2\pi}\int_0^{2\pi} f(\theta)\mathrm{d}\theta$, or $\frac{1}{2\pi}\int_{-\pi}^{\pi} f(\theta)\mathrm{d}\theta$, or $\frac{1}{2\pi}\int_{\theta\in S^1} f(\theta)\mathrm{d}\theta$; one identifies $L^p(S^1, \mathbb{R}^m)$ with the subset of $L^p(\mathbb{R}, \mathbb{R}^m)$ made up of 2π-periodic functions.

2.5. The Euclidean norm in $\mathbb{R}^m$ or $(\mathbb{R}^m)^*$ is denoted by $\|.\|$, and the ball of radius one centered is at the origin by B^m. We view an element of $\mathbb{R}^m$ as an $m \times 1$ matrix (column) of real numbers and an element of $(\mathbb{R}^m)^*$ as a $1 \times m$ matrix (line); transposition, denoted $.^\top$, sends $\mathbb{R}^m$ to $(\mathbb{R}^m)^*$ and vice versa.

3. Fast-oscillating control systems. We call *fast-oscillating control system* on M a family of nonautonomous systems, indexed by a positive number ε, linear in the control $u \in \mathbb{R}^m$:

$$\dot{x} = \mathcal{G}\left(\frac{t}{\varepsilon}, x\right) u = \sum_{i=1}^{m} \mathcal{G}_i\left(\frac{t}{\varepsilon}, x\right) u_i\,, \quad \|u\| \le 1. \tag{1}$$

Each $\mathcal{G}_i$ is a smooth "periodic time-varying" vector field: $\mathcal{G}_i \in C^\infty(S^1 \times M, \mathrm{T}M)$. An admissible control is a measurable $u(.) : [0,T] \to B^m$ for some $T > 0$. For a given control $u(.)$ and initial condition $x(0)$, there is a unique solution $x(.)$, defined either on $[0,T]$ or on a maximal interval $[0,T')$, $T' < T$.

Remark 3.1. Apart from being a notation defined by the double equality in (1), $\mathcal{G}(\theta, x)$ defines a linear map $\mathbb{R}^m \to \mathrm{T}_xM$ that sends $(u_1, \ldots, u_m)^\top$ to $\sum_{i=1}^m \mathcal{G}_i(\theta, x)u_i$.

3.1. Average control system of fast-oscillating control systems. Define the map $\overline{\mathcal{G}} : M \times L^\infty([0,2\pi], \mathbb{R}^m) \to \mathrm{T}M$ by

$$\overline{\mathcal{G}}(x, \mathcal{U}) = \frac{1}{2\pi}\int_0^{2\pi} \mathcal{G}(\theta, x)\, \mathcal{U}(\theta)\, \mathrm{d}\theta. \tag{2}$$

It allows one to define, for all $x \in M$, the subset $\mathcal{E}(x) \subset \mathrm{T}_xM$ by

$$\mathcal{E}(x) = \left\{\overline{\mathcal{G}}(x, \mathcal{U}),\, \mathcal{U} \in L^\infty\left([0,2\pi], \mathbb{R}^m\right),\, \|\mathcal{U}\|_\infty \le 1\right\} \subset T_xM, \tag{3}$$

and the *average control system* of (1) as follows.[1]

DEFINITION 3.2. *The average control system of* (1) *is the differential inclusion*

$$\dot{x} \in \mathcal{E}(x). \tag{4}$$

A solution of (4) *is an absolutely continuous* $x(.) : [0,T] \to M$ *such that* $\dot{x}(t) \in \mathcal{E}(x(t))$ *for almost all* t.

PROPOSITION 3.3. *For all* x *in* M, $\mathcal{E}(x)$ *is convex, compact, and symmetric with respect to the origin.*

[1] Its relation to the limit case of (1) as $\varepsilon \to 0$ is discussed in the next section.

Proof. $\mathcal{E}(x)$ is closed, convex, and symmetric because it is the image of the unit ball of $L^\infty\left(S^1, \mathbb{R}^m\right)$ by a linear map; it is compact because $\mathcal{G}(x,.)$ is bounded on S^1. ∎

Further characterizations of $\mathcal{E}(x)$ use the map $H : \mathrm{T}^*M \to [0,+\infty)$ defined by

$$H(x,p) = \frac{1}{2\pi}\int_0^{2\pi} \|\langle p, \mathcal{G}(\theta,x)\rangle\| \mathrm{d}\theta, \tag{5}$$

where the Euclidean norm is used according to section 2.5 and, for each (θ, x),

$$\langle p, \mathcal{G}(\theta,x)\rangle = (\langle p, \mathcal{G}_1(\theta,x)\rangle, \ldots, \langle p, \mathcal{G}_m(\theta,x)\rangle) \in (\mathbb{R}^m)^*\,. \tag{6}$$

PROPOSITION 3.4. *For all $(x,p) \in T^*M$, one has, with H defined as in* (5),

$$\mathcal{E}(x) = \left\{ v \in \mathrm{T}_xM\,,\ \sup_{\substack{p\in \mathrm{T}_x^*M \\ H(x,p)\le 1}} \langle p, v\rangle \ \le\ 1 \right\}, \tag{7}$$

$$H(x,p) \ = \sup_{v\in\mathcal{E}(x)} \langle p, v\rangle \ = \sup_{\mathcal{U}\in L^\infty(S^1,\mathbb{R}^m),\ \|\mathcal{U}\|_\infty\le 1} \langle p, \overline{\mathcal{G}}(x,\mathcal{U})\rangle \ = \ \langle p, \overline{\mathcal{G}}(x, \mathcal{U}^*_{p,x})\rangle \tag{8}$$

with $\mathcal{U}^*_{p,x} \in L^\infty\left(S^1, \mathbb{R}^m\right)$ *defined by*

$$\mathcal{U}^*_{p,x}(\theta) = \begin{cases} 0 & \text{if } \langle p, \mathcal{G}(\theta,x)\rangle = 0\,, \\ \frac{\langle p, \mathcal{G}(\theta,x)\rangle^\top}{\|\langle p, \mathcal{G}(\theta,x)\rangle\|} & \text{if } \langle p, \mathcal{G}(\theta,x)\rangle \neq 0\,. \end{cases} \tag{9}$$

Proof. The last equality in (8) is a straightforward maximization, the second one comes from the definition (3) of $\mathcal{E}(x)$, and a simple computation yields $H(x,p) = \langle p, \overline{\mathcal{G}}(x, \mathcal{U}^*_{p,x})\rangle$; this proves (8). Being closed and convex, $\mathcal{E}(x)$ is the intersection of all its supporting half-spaces [24, Corollary 1.3.5]; according to (8), this yields the following relation, equivalent to (7): $\mathcal{E}(x) = \bigcap_{p\in \mathrm{T}_x^*M} \{v \in \mathrm{T}_xM\,,\ \langle p, v\rangle \le H(x,p)\}$. ∎

A convenient characterization of solutions of (4). According to Definition 3.2, a solution $x(.)$ is such that, for almost all t, there is $\mathcal{U}(t) \in L^\infty([0,2\pi], \mathbb{R}^m)$ such that $\dot{x}(t) = \overline{\mathcal{G}}(x(t), \mathcal{U}(t))$; the map $(t,\theta) \mapsto \mathcal{U}(t)(\theta)$ is measurable with respect to θ only. It turns out that it may always be chosen jointly measurable with respect to (t,θ) according to the following "measurable selection" result.

PROPOSITION 3.5. *A map $x : [0,T] \to \mathbb{R}^n$ is a solution of the differential inclusion* (4) *if and only if there exists $\widehat{u} \in L^\infty([0,T]\times S^1, \mathbb{R}^m)$, $\|\widehat{u}\|_\infty \le 1$, such that*

$$\dot{x}(t) = \frac{1}{2\pi}\int_0^{2\pi} \mathcal{G}(\theta, x(t))\, \widehat{u}(t,\theta)\, \mathrm{d}\theta \tag{10}$$

for almost all t in $[0,T]$.

Proof. After possibly partitioning $[0,T]$ into intervals where $\dot{x}(t)$ remains in the same coordinate chart, we work in coordinates and use a Euclidean norm when useful.

Sufficiency is clear: from Fubini's theorem, $\theta \mapsto \widehat{u}(t,\theta)$ is measurable for almost all t, and hence $x(.)$ is a solution of (4). Conversely, let $x(.)$ be a solution of (4): $\dot{x}(.)$ is measurable, and, for almost all t, there exists $\tilde{u}_t \in L^\infty(S^1, \mathbb{R}^m)$, $\|\tilde{u}_t\|_\infty \le 1$, such that

$$\dot{x}(t) = \overline{\mathcal{G}}(x(t), \tilde{u}_t) = \frac{1}{2\pi}\int_0^{2\pi} \mathcal{G}(s_1, x(t))\tilde{u}_t(s_1)\mathrm{d}s_1\,. \tag{11}$$

Let $\phi : L^\infty\left([0,T]\times S^1,\mathbb{R}^m\right)\to L^2\left([0,T],\mathbb{R}^n\right)$ be the linear map defined by

$$\phi(u)(t)=\overline{\mathcal{G}}(x(t),u(t,.))=\frac{1}{2\pi}\int_0^{2\pi}\mathcal{G}(s_1,x(t))\,u(t,s_1)\,\mathrm{d}s_1,$$

and let $\mathcal{I}$ be the image by ϕ of the unit ball of $L^\infty([0,T]\times S^1,\mathbb{R}^m)$. Since, by (11), $\dot{x}(.)$ is essentially bounded, it is in $L^2\left([0,T],\mathbb{R}^n\right)$; since $\mathcal{I}$ is closed and convex in that Hilbert space, the distance from $\dot{x}$ to $\mathcal{I}$ is reached for a unique element $\bar{\xi}\in\mathcal{I}$:

$$\bar{\xi}=\phi(\bar{u})\,,\quad \bar{u}\in L^\infty\left([0,T]\times S^1,\mathbb{R}^m\right),\quad \|\bar{u}\|_{L^\infty}\le 1\,.$$

Let us prove by contradiction that $\bar{\xi}=\dot{x}$, i.e., $\dot{x}(.)\in\mathcal{I}$; this will end the proof.

If $\dot{x}\neq\bar{\xi}$, one has, for all u in the unit ball of $L^\infty\left([0,T]\times S^1,\mathbb{R}^m\right)$,

$$\left(\dot{x}-\bar{\xi}\,\middle|\,\phi(u)-\phi(\bar{u})\right)_{L^2}\;\le\;0, \tag{12}$$

with equality only if $\phi(u)=\phi(\bar{u})$. Define $\widehat{u}$ by $\widehat{u}(t,s)=\mathcal{U}^*_{\left(\dot{x}(t)-\bar{\xi}(t)\right)^\top,\,x(t)}(s)$, with $\mathcal{U}^*_{p,x}$ defined by (9); clearly, $\widehat{u}$ is in the unit ball of $L^\infty\left([0,T]\times S^1,\mathbb{R}^m\right)$, and, for all $(t,s)\in[0,T]\times S^1$ and all $\mathbf{u}\in\mathbb{R}^m$,

$$\|\mathbf{u}\|\le 1\quad\Rightarrow\quad\left(\dot{x}(t)-\bar{\xi}(t)\right)^\top\mathcal{G}(s,x(t))\;(\widehat{u}(t,s)-\mathbf{u})\ge 0, \tag{13}$$

and hence $\left(\dot{x}(t)-\bar{\xi}(t)\right)^\top\mathcal{G}(s_1,x(t))\;(\widehat{u}(t,s_1)-\bar{u}(t,s_1))$ is nonnegative for almost all (t,s_1), and, since it is the integrand of the left-hand side of (12), it must be zero; hence $\bar{\xi}=\phi(\bar{u})=\phi(\widehat{u})$ and $\bar{\xi}(t)=\overline{\mathcal{G}}(x(t),\widehat{u}(t,.))$ for almost all t.

In (11), $\tilde{u}_t$ satisfies $\|\tilde{u}_t(s_1)\|\le 1$ for almost all s_1; hence, according to (13),

$$\left(\dot{x}(t)-\bar{\xi}(t)\right)^\top\mathcal{G}(s_1,x(t))\;(\widehat{u}(t,s_1)-\tilde{u}_t(s_1))\ge 0\,.$$

Since $\dot{x}(t)=\overline{\mathcal{G}}(x(t),\tilde{u}_t)$, $\bar{\xi}(t)=\overline{\mathcal{G}}(x(t),\widehat{u}(t,.))$, the integration with respect to the variable s_1 yields $-\|\dot{x}(t)-\bar{\xi}(t)\|^2\ge 0$ for almost all t; this contradicts $\dot{x}\neq\bar{\xi}$. ∎

Remark 3.6. The differential inclusion (4) is equivalent to the "control system"

$$\dot{x}=\overline{\mathcal{G}}(x,\mathcal{U})\,,\quad \mathcal{U}\in L^\infty(S^1,\mathbb{R}^m)\,,\ \|\mathcal{U}\|_\infty\le 1,$$

where, by Proposition 3.5, admissible controls are maps $t\mapsto\mathcal{U}(t)$ such that $\widehat{u}\colon(t,\theta)\mapsto\mathcal{U}(t)(\theta)$ is measurable with respect to (t,θ). Since this "control" is infinite dimensional, and we could not find a representation of the type $\dot{x}=f(x,v)$, $v\in U\subset\mathbb{R}^r$, r finite, we stay with the differential inclusion (4), with $\mathcal{E}(x)$ described by (8) and (5).

3.2. Convergence theorem. The following result relates solutions of the fast-oscillating systems as ε tends to zero to solutions of the average system. To the best of our knowledge, this kind of theorem, where the control is not chosen prior to averaging, has never been stated in the literature.

THEOREM 3.7 (convergence for fast-oscillating control systems).

1. *Let $x_0(.):[0,T]\to M$ be an arbitrary solution of (4). There exist a family of measurable functions $\overline{u}_\varepsilon(.):[0,T]\to B^m$, indexed by $\varepsilon>0$, and positive constants c,ε_0 such that, calling $x_\varepsilon(.)$ the solution of (1) with control $u=\overline{u}_\varepsilon(t)$ and initial condition $x_\varepsilon(0)=x_0(0)$, one has that $x_\varepsilon(.)$ is defined on $[0,T]$ for all ε smaller than ε_0 and converges to $x_0(.)$ as $\varepsilon\to 0$, with an error of uniform order ε:*

$$d(x_\varepsilon(t),x_0(t))<c\,\varepsilon\,,\quad t\in[0,T]\,,\ 0<\varepsilon<\varepsilon_0\,. \tag{14}$$

2. *Let $\mathbb{K}$ be a compact subset of M, let $(\varepsilon_n)_{n\in\mathbb{N}}$ be a decreasing sequence of positive real numbers converging to zero, and, for each n, let $u_n(.) \in L^\infty([0,T],\mathbb{R}^m)$, $\|u_n(.)\|_\infty \le 1$, be a control and $x_n(.) : [0,T] \to \mathbb{K}$ be the solution of* (1) *with $\varepsilon = \varepsilon_n$ and $u = u_n(t)$. Then the sequence $\big(x_n(.)\big)_{n\in\mathbb{N}}$ is compact for the topology of uniform convergence on $[0,T]$ and any accumulation point is a solution of the average system* (4).

The statement is more complex than that for ODEs (see, e.g., [2, sect. 52.C] due to underdetermination (choice of control in (1), and multivalued right-hand side in (4)).

Informally, "1" states that any solution of the average system is the limit of solutions of fast-oscillating systems with well chosen controls, and "2" states that, conversely, any limit of solutions of oscillating systems, with arbitrary controls, is a solution of the average control system. There is an estimate on the error in "1" but not in "2" because some sequences may converge more slowly than others.

Remark 3.8. One may consider systems that are *affine* instead of linear in the control by adding a drift vector field $\mathcal{G}_0(t/\varepsilon, x)$ to (1). Then, in the average control system, $\mathcal{E}(x)$ is replaced by $\overline{\mathcal{G}}_0(x) + \mathcal{E}(x)$, with $\overline{\mathcal{G}}_0(x) = \frac{1}{2\pi}\int_0^{2\pi} \mathcal{G}_0(\theta, x)\mathrm{d}\theta$. By a straightforward extension, convergence holds for these systems too.

In the proof of Theorem 3.7, the following technical lemma is needed.

Lemma 3.9. *Let $\varepsilon > 0$ and $a < b$ be real numbers, and let $\widehat{u} : [a - 2\pi\varepsilon, b] \times S^1 \to \mathbb{R}^m$ be measurable. One has the following identity (see section* 2.4 *for the notation $\mu(.)$):*

$$\iint_{\substack{\theta\in S^1\\ a\le s\le b}} \mathcal{G}(\theta, x(s))\, \widehat{u}(s,\theta)\, \mathrm{d}\theta\, \mathrm{d}s = \iint_{\substack{\theta\in S^1\\ a\le s\le b}} \mathcal{G}\left(\frac{s}{\varepsilon}, x(s)\right) \widehat{u}\left(s + \varepsilon\mu(\theta), \frac{s}{\varepsilon}\right) \mathrm{d}\theta\, \mathrm{d}s + \Delta_\varepsilon \tag{15}$$

with

$$\begin{aligned}\Delta_\varepsilon = &\iint_{T_\varepsilon^a} \mathcal{G}\left(\frac{s}{\varepsilon}, x(s+\varepsilon\mu(\theta))\right) \widehat{u}\left(s+\varepsilon\mu(\theta), \frac{s}{\varepsilon}\right) \mathrm{d}\theta\, \mathrm{d}s \\ &- \iint_{T_\varepsilon^b} \mathcal{G}\left(\frac{s}{\varepsilon}, x(s+\varepsilon\mu(\theta))\right) \widehat{u}\left(s+\varepsilon\mu(\theta), \frac{s}{\varepsilon}\right) \mathrm{d}\theta\, \mathrm{d}s \\ &+ \iint_{\substack{\theta\in S^1\\ a\le s\le b}} \left[\mathcal{G}\left(\frac{s}{\varepsilon}, x(s+\varepsilon\mu(\theta))\right) - \mathcal{G}\left(\frac{s}{\varepsilon}, x(s)\right)\right] \widehat{u}\left(s+\varepsilon\mu(\theta), \frac{s}{\varepsilon}\right) \mathrm{d}\theta\, \mathrm{d}s\end{aligned} \tag{16}$$

and the set T_ε^a defined by $T_\varepsilon^a = \{(s,\theta),\, \theta\in S^1,\, a - \varepsilon\,\mu(\theta) \le s \le a\}$ and T_ε^b accordingly.

Proof. Thanks to the change of variables $\theta = \tau/\varepsilon \bmod 2\pi$, $s = \tau + \varepsilon\,\mu(\phi)$, with $\mu(\theta)$ as defined in section 2.4, the left-hand side of (15) is equal to

$$\iint_{\substack{\phi\in S^1\\ a-\varepsilon\phi\le\tau\le b-\varepsilon\mu(\phi)}} \mathcal{G}\left(\frac{\tau}{\varepsilon}, x(\tau+\varepsilon\mu(\phi))\right) \widehat{u}\left(\tau+\varepsilon\mu(\phi), \frac{\tau}{\varepsilon}\right) \mathrm{d}\tau\, \mathrm{d}\phi\ .$$

Keeping the names (s,θ) instead of (τ,ϕ), one gets (15), with the correcting term Δ_ε coming from the modified domain of integration and argument of x. ∎

Proof of Theorem 3.7, *point* 1. Consider a solution $x_0 : [0,T] \to M^n$ of (4). By Proposition 3.5 there exists $\widehat{u}_0 \in L^\infty([0,T]\times S^1, \mathbb{R}^m)$, $\|\widehat{u}_0\|_\infty \le 1$, satisfying (10). For $\varepsilon > 0$, define $\overline{u}_\varepsilon(.) \in L^\infty([0,T],\mathbb{R}^m)$ by (see section 2.4 for notation)

$$\overline{u}_\varepsilon(t) = \frac{1}{2\pi}\int_0^{2\pi} \widehat{u}_0\left(t + \varepsilon\,\mu(\theta), \frac{t}{\varepsilon}\right) \mathrm{d}\theta, \tag{17}$$

where $\widehat{u}_0$ is prolonged by zero outside $[0,T]$: $\widehat{u}_0(t+\varepsilon\,\mu(\theta),\frac{t}{\varepsilon})=0$ if $t+\varepsilon\,\mu(\theta)>T$. Let us prove that this construction of $\overline{u}_\varepsilon$ satisfies the two announced properties.

Step 1. *Let us first assume that M is an open subset of $\mathbb{R}^n$ and $\mathcal{G}$ is zero outside a compact subset of M.* Then $\mathcal{G}(\theta,x)$ is an $n\times m$ matrix for all (θ,x), and, denoting by $\|.\|$ the Euclidean norm for vectors and the operator norm for matrices, there are global constants $\operatorname{Lip}\mathcal{G}$ and $\sup\mathcal{G}$ such that, for all x,x',θ in $M\times M\times S^1$,

$$\|\mathcal{G}(\theta,x)-\mathcal{G}(\theta,x')\|\leq\big(\operatorname{Lip}\mathcal{G}\big)\|x-x'\|\,,\quad\|\mathcal{G}(\theta,x)\|\leq\big(\sup\mathcal{G}\big)\,. \tag{18}$$

Let b be a nonnegative constant and consider, for each $\varepsilon>0$, a solution $x_\varepsilon(.)$ of (1) with control $u=\overline{u}_\varepsilon(t)$ and initial condition $x_\varepsilon(0)$ such that

$$\|x_\varepsilon(0)-x_0(0)\|\leq b\,\varepsilon\,. \tag{19}$$

In fact $b=0$ in the theorem itself, but we need a nonzero b in Step 2. By definition, expanding $\overline{u}_\varepsilon(s)$ as in (17) and using Lemma 3.9, one has

$$\begin{aligned} x_\varepsilon(t) &= x_\varepsilon(0)+\frac{1}{2\pi}\int_0^t\int_0^{2\pi}\mathcal{G}\left(\frac{s}{\varepsilon},x_\varepsilon(s)\right)\widehat{u}_0\left(s-\varepsilon\,\mu(\theta),\frac{s}{\varepsilon}\right)\mathrm{d}\theta\,\mathrm{d}s\\ &= x_\varepsilon(0)+\frac{1}{2\pi}\left(\int_0^t\int_0^{2\pi}\mathcal{G}(\theta,x_\varepsilon(s))\,\widehat{u}_0(s,\theta)\,\mathrm{d}\theta\,\mathrm{d}s\;-\;\Delta_\varepsilon\right), \end{aligned} \tag{20}$$

with Δ_ε given by (16), that satisfies $\|\Delta_\varepsilon\|\leq 4\pi^2\big(\operatorname{Lip}\mathcal{G}\big)\,(1+T\sup\mathcal{G})\,\varepsilon$ because, in particular, $\|\widehat{u}_0\|\leq 1$, $|\varepsilon\mu(\theta)|<2\pi\varepsilon$, and

$$\left\|\left(\mathcal{G}\left(\frac{s}{\varepsilon},x_\varepsilon(s)\right)-\mathcal{G}\left(\frac{s}{\varepsilon},x_\varepsilon(s+\varepsilon\,\mu(\theta))\right)\right)\widehat{u}_0\left(s+\varepsilon\,\mu(\theta),\frac{s}{\varepsilon}\right)\right\|\leq 2\pi\,\big(\operatorname{Lip}\mathcal{G}\big)\,\big(\sup\mathcal{G}\big)\,\varepsilon\,.$$

Using (19), (20), the bound on $\|\Delta_\varepsilon\|$, and the relation

$$x_0(t)=x_0(0)+\frac{1}{2\pi}\int_0^t\int_0^{2\pi}\mathcal{G}(\theta,x_0(s))\,\widehat{u}_0(s,\theta)\,\mathrm{d}\theta\,\mathrm{d}s\,,$$

one gets

$$\|x_\varepsilon(t)-x_0(t)\|\leq\big(b+2\pi\,\big(\operatorname{Lip}\mathcal{G}\big)\,\big(1+T\sup\mathcal{G}\big)\big)\,\varepsilon+\big(\operatorname{Lip}\mathcal{G}\big)\int_0^t\|x_\varepsilon(s)-x_0(s)\|\,\mathrm{d}s$$

for all t in $[0,T]$, and, finally, by Gronwall's lemma,

$$\|x_\varepsilon(t)-x_0(t)\|\leq\big[b+2\pi\big(\operatorname{Lip}\mathcal{G}\big)\,(1+T\sup\mathcal{G})\big]\;e^{T\operatorname{Lip}\mathcal{G}}\,\varepsilon \tag{21}$$

for all t in $[0,T]$ and ε in $[0,\varepsilon_0]$. This proves the theorem if M is an open subset of $\mathbb{R}^n$ and $\mathcal{G}$ is zero outside a compact subset, with an explicit constant c corresponding to the distance d defined from the Euclidean norm and with $\varepsilon_0=+\infty$.

Step 2. *General case.* Let $x_\varepsilon(.)$ be the solution of (1) with control $u=\overline{u}_\varepsilon(t)$ defined in (17) from $\widehat{u}_0$ and with initial condition $x_\varepsilon(0)=x_0(0)$; it is not necessarily defined on $[0,T]$ but may have a maximum interval of definition $[0,T_\varepsilon)$ with $T_\varepsilon<T$. Let $\widetilde{T}\in[0,T]$ be the supremum of the set of numbers $\tau\in[0,T]$ such that, for some ε_0 and some c that may depend on τ, the solution $x_\varepsilon(.)$ is defined on $[0,\tau]$ and satisfies $d(x_\varepsilon(t),x_0(0))<c\,\varepsilon$ for all $t\in[0,\tau]$ and $\varepsilon\in[0,\varepsilon_0]$. Let us prove by contradiction that $\widetilde{T}=T$. This will end the proof of Theorem 3.7, point 1.

Assume $\widetilde{T} < T$, and let

- $\mathcal{O}$ be a coordinate neighborhood of $x_0(\widetilde{T})$,

- $\alpha > 0$ be such that $0 < \widetilde{T} - \alpha < \widetilde{T} + \alpha \leq T$ and $x_0([\widetilde{T} - \alpha, \widetilde{T} + \alpha]) \subset \mathcal{O}$,

- $c > 0$, $\varepsilon_0 > 0$ be such that $d(x_\varepsilon(t), x_0(t)) < c\,\varepsilon$ for all $t \in [0, \widetilde{T} - \alpha]$ and $\varepsilon \in [0, \varepsilon_0]$.

Taking ε_0 possibly smaller, one also has $x_\varepsilon(\widetilde{T} - \alpha) \in \mathcal{O}$ for $\varepsilon < \varepsilon_0$. Let $\mathbb{K}$ be a compact neighborhood of $x_0([\widetilde{T} - \alpha, \widetilde{T} + \alpha])$ contained in $\mathcal{O}$, let $\mathbb{K}'$ be a compact neighborhood of $\mathbb{K}$ contained in $\mathcal{O}$, and let $\rho : M \to [0, 1]$ be a smooth map, zero outside $\mathbb{K}'$, and constant equal to 1 in $\mathbb{K}$. Defining $\mathcal{G}_\rho$ by $\mathcal{G}_\rho(\theta, x) = \rho(x)\mathcal{G}(\theta, x)$, let us apply Step 1 in coordinates in $\mathcal{O}$, with $\mathcal{G}_\rho$ instead of $\mathcal{G}$ and $[\widetilde{T} - \alpha, \widetilde{T} + \alpha]$ instead of $[0, T]$. Call x_0^ρ (resp., x_ε^ρ, $\varepsilon > 0$) the solution of (10) (resp., of (1) with control $u = \overline{u}_\varepsilon(t)$), replacing $\mathcal{G}$ by $\mathcal{G}_\rho$, with initial condition $x_\varepsilon^\rho(\widetilde{T} - \alpha) = x_\varepsilon(\widetilde{T} - \alpha)$, $\varepsilon \geq 0$. One clearly has, as in (19), $\|x_\varepsilon^\rho(\widetilde{T} - \alpha) - x_0^\rho(\widetilde{T} - \alpha)\| < b\,\varepsilon$, with b deduced from c via Lipschitz equivalence of the distance d and the Euclidean norm in coordinates (see section 2.3); then Step 1 provides $\varepsilon_0' > 0$ such that, by (21), the inequality

$$\|x_\varepsilon^\rho(t) - x_0^\rho(t)\| \leq \left[b + 2\pi\big(\operatorname{Lip}\mathcal{G}_\rho\big)\left(1 + 2\alpha \sup \mathcal{G}_\rho\right)\right] e^{2\alpha \operatorname{Lip}\mathcal{G}_\rho}\,\varepsilon \tag{22}$$

is valid for $t \in [\widetilde{T} - \alpha, \widetilde{T} + \alpha]$ and $\varepsilon \in [0, \varepsilon_0']$. Possibly choosing a smaller ε_0', this implies that $x_\varepsilon([\widetilde{T} - \alpha, \widetilde{T} + \alpha]) \subset \mathbb{K}$ for $\varepsilon < \varepsilon_0'$; since $\mathcal{G}$ coincides with $\mathcal{G}_\rho$ in $\mathbb{K}$, the conclusion holds for x_ε and $\mathcal{G}$ as well as for x_ε^ρ and $\mathcal{G}_\rho$ if ε is no larger than ε_0'. We have shown that, for all $\varepsilon < \varepsilon_0'$, the solution x_ε is defined on $[0, \widetilde{T} + \alpha]$ and satisfies $d(x_\varepsilon(t), x_0(t)) \leq c'\varepsilon$ for t in $[0, \widetilde{T} + \alpha]$, where c' is larger than c and than a bound deduced from (22) and from Lipschitz equivalence of d with the Euclidean distance. This contradicts the definition of $\widetilde{T}$. ∎

Proof of Theorem 3.7, *point* 2. Since $\mathcal{G}$ is bounded on $S^1 \times \mathbb{K}$ (one may cover $\mathbb{K}$ with a finite number of coordinate charts and define this bound in coordinates), the maps $x_n(.)$ have a common Lipschitz constant, and the sequence $(x_n(.))$ is equicontinuous and hence compact by the Ascoli–Arzela theorem: one may extract a uniformly convergent subsequence. Still denoting by $(x_n(.))_{n\in\mathbb{N}}$ such a converging subsequence and by $x^*(.)$ its (uniform) limit, we need to prove that this limit is a solution of (4).

Define, for each n, $\widehat{u}_n : [0, T] \times S^1 \to \mathbb{R}^m$ by

$$\widehat{u}_n(t, \theta) = u_n(\,\beta_n(t, \theta)\,)\,, \tag{23}$$

where $u_n(.) \in L^\infty([0, T], \mathbb{R}^m)$ is associated to $x_n(.)$ according to the assumption of the theorem, and where the map $\beta_n : [0, T] \times S^1 \to \mathbb{R}$ is defined by

$$t - 2\pi\varepsilon_n < \beta_n(t, \theta) \leq t\,, \quad \frac{\beta_n(t, \theta)}{\varepsilon_n} \equiv \theta \mod 2\pi\,. \tag{24}$$

Clearly $\widehat{u}_n$ is in $L^\infty([0, T] \times S^1, \mathbb{R}^m)$ and $\|\widehat{u}_n\|_\infty \leq 1$. Hence, after possibly extracting a subsequence, $(\widehat{u}_n)$ converges in the weak-$*$ topology to some $\widehat{u}^*$. Let us prove that, for almost all $t \in [0, T]$,

$$\dot{x}^*(t) = \frac{1}{2\pi}\int_0^{2\pi} \mathcal{G}(\theta, x^*(t))\,\widehat{u}^*(t, \theta)\mathrm{d}\theta\,. \tag{25}$$

Let $\widetilde{T} \in [0, T]$ be the supremum of the set of numbers $\tau \in [0, T]$ such that this is true for almost all t in $[0, \tau]$, and let us prove by contradiction that $\widetilde{T} = T$.

Assume $\widetilde{T} < T$, and let $\mathcal{O}$ be a coordinate neighborhood of $x_0(\widetilde{T})$ and α be such that $0 < \widetilde{T} - \alpha < \widetilde{T} + \alpha \leq T$ and $x_0([\widetilde{T} - \alpha, \widetilde{T} + \alpha]) \subset \mathcal{O}$. Uniform convergence implies $x_n([\widetilde{T} - \alpha, \widetilde{T} + \alpha]) \subset \mathcal{O}$ for n large enough and then, in coordinates, for $t \in [\widetilde{T} - \alpha, \widetilde{T} + \alpha]$,

$$x_n(t) - x_n(\widetilde{T} - \alpha) = \int_{\widetilde{T}-\alpha}^{t} \mathcal{G}\left(\frac{s}{\varepsilon_n}, x_n(s)\right) u_n(s)\,\mathrm{d}s\,. \tag{26}$$

From (24), one has $\beta_n(s + \varepsilon_n\theta, \frac{s}{\varepsilon_n}) = s$; hence, from (23), $\widehat{u}_n(s + \varepsilon_n\theta, \frac{s}{\varepsilon_n}) = u_n(s)$ for all $\theta \in S^1$, $s \in \mathbb{R}$; using this in Lemma 3.9, one has

$$\frac{1}{2\pi} \iint_{\substack{\widetilde{T}-\alpha \leq s \leq t \\ \theta \in S^1}} \mathcal{G}(\theta, x_n(s)) \widehat{u}_n(s, \theta)\,\mathrm{d}\theta\,\mathrm{d}s = \frac{1}{2\pi} \iint_{\substack{\widetilde{T}-\alpha \leq s \leq t \\ \theta \in S^1}} \mathcal{G}\left(\frac{s}{\varepsilon_n}, x_n(s)\right) u_n(s)\,\mathrm{d}\theta\,\mathrm{d}s \;+ \Delta_{\varepsilon_n}\,.$$

Since the integral in the right-hand side—whose integrand does not depend on θ—is equal to the right-hand side of (26), one gets, using uniform convergence of x_n to x^*, weak convergence of $\widehat{u}_n$ to $\widehat{u}^*$, and convergence of Δ_{ε_n} to zero,

$$x^*(t) - x^*(\widetilde{T} - \alpha) = \frac{1}{2\pi} \int_{\widetilde{T}-\alpha}^{t} \int_0^{2\pi} \mathcal{G}(\theta, x^*(s)) \widehat{u}^*(s, \theta)\,\mathrm{d}\theta\,\mathrm{d}s$$

for t in $[\widetilde{T} - \alpha, \widetilde{T} + \alpha]$, and finally that (25) hold for almost all t in $[0, \widetilde{T} + \alpha]$, thus contradicting the definition of $\widetilde{T}$. ∎

3.3. Dimension of the velocity set $\mathcal{E}(x)$. Recall that, for a convex subset C of a linear space containing the origin, its linear hull is the smallest linear subspace that contains C, the interior of C *in its linear hull* is always nonempty, and $\dim C$ is the dimension of this linear hull.

Viewing $\frac{\partial^j \mathcal{G}}{\partial \theta^j}(\theta, x)$ as a linear map $\mathbb{R}^m \to \mathrm{T}_x M$ (see Remark 3.1), and denoting by Σ a sum of linear subspaces of $T_x M$, define the integer $r(\theta, x)$ by

$$r(\theta, x) = \dim\left(\sum_{j \in \mathbb{N}} \mathrm{Range}\, \frac{\partial^j \mathcal{G}}{\partial \theta^j}(\theta, x)\right). \tag{27}$$

It is also the rank of the collection of vectors $\frac{\partial^j \mathcal{G}_i}{\partial \theta^j}(\theta, x) \in \mathrm{T}_x M$, $1 \leq i \leq m$, $j \geq 0$.

In the following proposition—the sole place where this property is used—"system (1) is real analytic with respect to θ" means that the vector fields $\mathcal{G}_i$ are real analytic with respect to θ for fixed x (while being smooth with respect to (θ, x)).

PROPOSITION 3.10.

1. *The linear hull of $\mathcal{E}(x)$ satisfies the following two properties for all x in M, where the inclusion* (29) *is an equality if* (1) *is real analytic with respect to θ:*

$$\text{Linear hull}\, \mathcal{E}(x) \;=\; \sum_{\theta \in S^1} \mathrm{Range}\, \mathcal{G}(\theta, x)\,, \tag{28}$$

$$\text{Linear hull}\, \mathcal{E}(x) \;\supset\; \sum_{j \in \mathbb{N}} \mathrm{Range}\, \frac{\partial^j \mathcal{G}}{\partial \theta^j}(\theta, x) \qquad \textit{for all } \theta \in S^1\,. \tag{29}$$

2. *If $r(\theta, x) = n$ for at least one θ in S^1, then $\mathcal{E}(x)$ has a nonempty interior in $T_x M$, i.e., $\dim \mathcal{E}(x) = n$.*

3. *If the system* (1) *is real analytic with respect to θ, then $r(\theta, x)$ does not depend on θ and $r(\theta, x) = \dim \mathcal{E}(x)$.*

Proof. If p is in $\operatorname{Range}\mathcal{G}(\theta,x)^{\perp}$ for all θ, then any $v=\overline{\mathcal{G}}(x,\mathcal{U})$ in $\mathcal{E}(x)$ satisfies $\langle p,v\rangle=0$ because $\langle p,\mathcal{G}(\theta,x)\mathcal{U}(\theta)\rangle$ is identically zero on $[0,2\pi]$. Conversely, let p be in $\mathcal{E}(x)^{\perp}$, and consider $v=\overline{\mathcal{G}}(x,\mathcal{U}^*_{p,x})\in\mathcal{E}(x)$; then $\langle p,v\rangle=0$ implies $\langle p,\mathcal{G}(\theta,x)\rangle=0$, i.e., $p\in\operatorname{Range}\mathcal{G}(\theta,x)^{\perp}$ for all θ. We have proved the identity $\mathcal{E}(x)^{\perp}=\bigcap_{\theta\in S^1}(\operatorname{Range}\mathcal{G}(\theta,x))^{\perp}$, and hence proved (28).

If p is in $\operatorname{Range}\mathcal{G}(\phi,x)^{\perp}$ for all ϕ, differentiating $\langle p,\mathcal{G}(\phi,x)\rangle=0$ with respect to ϕ yields $\left\langle p,\partial^j\mathcal{G}/\partial\phi^j(x,\phi)\right\rangle=0$, $j\in\mathbb{N}$; we have proved, taking $\phi=\theta$, the inclusion $\bigcap_{\phi\in S^1}(\operatorname{Range}\mathcal{G}(\phi,x))^{\perp}\subset\bigcap_{j\in\mathbb{N}}\left(\operatorname{Range}\frac{\partial^j\mathcal{G}}{\partial\theta^j}(\theta,x)\right)^{\perp}$, and hence proved (29). To prove the reverse inclusion in the real analytic case, fix $\theta\in S^1$ and $p\in\bigcap_{j\in\mathbb{N}}\frac{\partial^j\mathcal{G}}{\partial\theta^j}(\theta,x)^{\perp}$, and consider the real analytic mapping $S^1\to(\mathbb{R}^m)^*$, $\phi\mapsto\langle p,\mathcal{G}(\phi,x)\rangle$; the assumption on p implies that this map vanishes for $\phi=\theta$, as well as its derivatives at all orders, and hence it is identically zero: $p\in\bigcap_{\phi\in S^1}(\operatorname{Range}\mathcal{G}(\phi,x))^{\perp}$. This ends the proof of point 1. Point 2 is an easy consequence, and point 3 is classical. ∎

3.4. Further properties in the full rank case. We now assume that the mapping $\mathcal{G}$ in (1) is such that the rank $r(\theta,x)$ defined by (27) is maximal:

$$r(\theta,x)=n \text{ for all } x \text{ in } M \text{ and } \theta \text{ in } S^1. \tag{30}$$

3.4.1. Controllability. Condition (30) is strongly related to controllability of the linear approximation of (1) around equilibria, i.e., around solutions where x is constant and u is identically zero. Indeed, the linear approximation of the time-varying nonlinear system (take $\varepsilon=1$ in (1))

$$\dot{x}=\mathcal{G}(t,x)u \tag{31}$$

around the equilibrium $x=x_1$ is the time-varying linear system $\dot{\xi}=\mathcal{G}(t,x_1)u$; according to [16, p. 614], it is "controllable with impulsive controls at any time" if and only if $r(t,x_1)=n$ for all t. If this is true at all points x_1, then all end-point mappings are submersions around zero controls; we shall need the following more precise result.

PROPOSITION 3.11. *Assume that* (30) *holds.*

1. *For all $x_1\in M$ and $T>0$, there exist a coordinate neighborhood $\mathcal{W}$ of x_1 (the ball $\mathcal{B}$ below refers to the Euclidean norm in these coordinates), positive constants α_0,c_3, and, for all $y\in\mathcal{W}$, a smooth map $\chi_y:\mathcal{B}(y,\alpha_0)\to L^\infty([0,T],\mathbb{R}^m)$ with Lipschitz constant c_3, which is a right inverse of the end-point mapping of* (31) *on $[0,T]$ starting from y; i.e., for all $y_f\in\mathcal{B}(y,\alpha_0)$, the control $\chi_y(y_f):[0,T]\to\mathbb{R}^m$ is such that the solution of $\dot{x}=\mathcal{G}(t,x)\chi_y(y_f)(t)$, $x(0)=y$ satisfies $x(T)=y_f$.*
2. *For all $\varepsilon>0$, the system* (1) *is fully controllable; i.e., there exists, for any $\varepsilon>0$ and any two point x_0,x_1 in M, a time T and a measurable control $u:[0,T]\to B^m$ such that the solution of* (1) *with $x(0)=x_0$ satisfies $x(T)=x_1$.*

Proof. Let $E_y:L^\infty([0,T],\mathbb{R}^m)\to M$ be the end-point mapping with starting point y. Condition (30) implies that the derivative of E_{x_1} at the zero control has rank n; hence there exists an n-dimensional subspace V of $L^\infty([0,T],\mathbb{R}^m)$ such that the restriction of E_{x_1}, and hence of E_y for y close enough, to V is a local diffeomorphism at zero; the χ_y's are the local inverses of these local diffeomorphisms: they depend smoothly on y, and hence α_0 and c_3 may be taken independent of y in point 1.

This implies that the reachable set from any point at any positive or negative time contains a neighborhood of this point; a classical argument then tells us that the reachable set from a point x_0 is M, assumed to be connected (see section 2.1), for it is both open (obvious) and closed (if $\bar{x}$ is in the closure of the reachable set, some points in the reachable set can be reached in negative time, and hence $\bar{x}$ can be reached from x_0). ◻

Let us now turn to the average system (4). From $H : \mathrm{T}^*M \to [0, +\infty)$ defined by (5), we define $N : \mathrm{T}M \to [0, +\infty]$ by

$$N(x, v) = \max_{p \in \mathrm{T}_x^*M,\ H(x,p) \le 1} \langle p, v \rangle \ . \tag{32}$$

Proposition 3.12. *Assume the rank condition* (30)*.*

1. *For all $x \in M$, $H(x, .)$ defines a norm on the cotangent space T_x^*M, its dual norm on the tangent space T_xM is $N(x, .)$, and $\mathcal{E}(x)$ is the unit ball for $N(x, .)$, i.e., $\mathcal{E}(x) = \{v \in \mathrm{T}_xM,\ N(x, v) \le 1\}$.*
2. *System* (4) *is fully controllable; i.e., there exists, for any points x_0, x_1 in M, a time T and a solution $x(.) : [0, T] \to M$ of* (4) *such that $x(0) = x_0$, $x(T) = x_1$.*

Proof. From (5), $H(x, p) = 0$ implies $\langle p, \mathcal{G}(\theta, x)\rangle = 0$ for all θ and, differentiating with respect to θ and using (30), this implies $p = 0$; this makes $p \mapsto H(x, p)$ a norm, and the other properties are straightforward. Hence N given by (32) is finite for any (x, v) and is, by definition, the dual norm of $H(x, .)$; $\mathcal{E}(x)$ is its unit ball by (7) in Proposition 3.4. To prove point 2, take a continuously differentiable curve $\gamma : [0, 1] \to M$ such that $\gamma(0) = x_0$, $\gamma(1) = x_1$, and $\sigma : [0, T] \to [0, 1]$ for some $T > 0$ such that

$$t \ge \int_0^{\sigma(t)} N\left(\gamma(s), \frac{\mathrm{d}\gamma}{\mathrm{d}s}(s)\right)\mathrm{d}s$$

(N and H are obviously continuous); then $t \mapsto x(t) = \gamma(\sigma(t))$ is a solution of (4) such that $x(0) = x_0$ and $x(T) = x_1$. ◻

3.4.2. On the differentiability of *H*. It is clear that H, given by (5), is as smooth as $\mathcal{G}$ on $\mathrm{T}^*M \setminus \widetilde{\mathcal{Z}}$ with

$$\widetilde{\mathcal{Z}} = \{(x, p) \in \mathrm{T}^*M\ \exists \theta \in S^1,\ \langle p, \mathcal{G}(\theta, x)\rangle = 0\} \ . \tag{33}$$

Unfortunately, $\widetilde{\mathcal{Z}}$ is not empty in general: it is generically a $(2n - m + 1)$-dimensional submanifold of T^*M. However, one has the following result, valid also at these points.

Theorem 3.13. *If condition* (30) *holds, H^2 is continuously differentiable.*

This theorem is stated for $H^2 : (x, p) \mapsto H(x, p)^2$, because H itself, homogeneous of degree 1 with respect to p, cannot be differentiable on $\{(x, p) \in T^*M, p = 0\}$, the set of points where H cancels according to Proposition 3.12, item 1.

The map H fails in general to be twice differentiable on $\widetilde{\mathcal{Z}}$. We have the following estimate of the modulus of continuity of its first derivative, that even fails to be Lipschitz continuous. Its main consequence is Theorem 3.21.

Theorem 3.14. *Assume that the rank condition* (30) *holds and that*

(i) *for $(x, p) \in \mathrm{T}^*M$, $p \ne 0$, there is at most one $\theta \in S^1$ such that $\langle p, \mathcal{G}(\theta, x)\rangle = 0$, and $\langle p, \frac{\partial \mathcal{G}}{\partial \theta}(\theta, x)\rangle$ does not vanish at the same point;*

(ii) *for all $(\theta, x) \in S^1 \times M$, one has $\operatorname{rank} \mathcal{G}(\theta, x) = m$;*

then any point $(\bar{x}, \bar{p})$ has a constant c and a coordinate neighborhood in T^*M such that for all X and Y in $\mathbb{R}^{2n}$, coordinates of points in the neighborhood,

$$\|\mathrm{d}H(Y) - \mathrm{d}H(X)\| \le c\,\|X - Y\|\,\ln \frac{1}{\|X - Y\|} \ . \tag{34}$$

In the left-hand side, $\|.\|$ stands for the operator norm in coordinates; see section 2.3.

Remark 3.15 (Finsler geometry). If H^2 was an least *twice* continuously differentiable, with a positive definite Hessian with respect to p, then so would N^2 (see (32)), and it would define a (reversible) *Finsler metric* [3] on M. The lack of differentiability calls for further developments.

Before proving these theorems, we state a more generic result, whose notation is totally independent from the rest of the paper. Its proof is in the appendix.

PROPOSITION 3.16. *Let* d *be a positive integer,* O^{d} *an open subset of* $\mathbb{R}^{\mathsf{d}}$*,* $\mathsf{V}\colon S^1 \times O^{\mathsf{d}} \to \mathbb{R}^m$ *a smooth map* (C^∞)*,* $\widetilde{\mathcal{Z}}$ *the subset of* O^{d} *where* V *vanishes for some* θ*, and* $\mathsf{H}\colon O^{\mathsf{d}} \to [0,+\infty)$ *the average of the norm of* V*:*

$$\mathsf{H}(X) = \frac{1}{2\pi}\int_0^{2\pi} \|\mathsf{V}(\theta,X)\|\,\mathrm{d}\theta\ , \quad \widetilde{\mathcal{Z}} = \{X \in O^{\mathsf{d}}\ \exists\theta \in S^1,\ \mathsf{V}(\theta,X)=0\}\ . \tag{35}$$

1. *Assume that, for all* X *in* O^{d}*,*

$$\textit{the set } \{\theta \in S^1\ ,\ \mathsf{V}(\theta,X)=0\} \textit{ has measure zero in } S^1. \tag{36}$$

Then H *is continuously differentiable and, for all* X*,*

$$\mathrm{d}\mathsf{H}(X).h = \frac{1}{2\pi}\int_0^{2\pi} \left(\frac{\partial \mathsf{V}}{\partial X}(\theta,X).h \,\middle|\, \frac{\mathsf{V}(\theta,X)}{\|\mathsf{V}(\theta,X)\|} \right) \mathrm{d}\theta\ . \tag{37}$$

2. *Let* V *satisfy the following assumptions:*

$$\begin{array}{ll} \text{(a)} & \mathsf{V}(\theta,X)=0 \textit{ for all } (\theta,X)\in S^1\times O^{\mathsf{d}} \textit{ such that } \mathsf{V}(\theta,X)=0;\\ \text{(b)} & \textit{for any } X\in O^{\mathsf{d}}\textit{, there is at most one } \theta \textit{ such that } \mathsf{V}(\theta,X)=0;\\ \text{(c)} & \mathsf{V} \textit{ and } \partial\mathsf{V}/\partial\theta \textit{ do not vanish simultaneously;} \end{array} \tag{38}$$

and let $\bar{X}$ *be in* $\widetilde{\mathcal{Z}}$*. There is a neighborhood* U *of* $\bar{X}$ *in* O^{d}*, and a constant* $K>0$ *such that, for all* X, Y*, in* U*,*

$$\Big\|\mathrm{d}\mathsf{H}(X) - \mathrm{d}\mathsf{H}(Y)\Big\| \le K\,\|X-Y\|\,\ln\frac{1}{\|X-Y\|}\ . \tag{39}$$

In the left-hand side of (39), $\|.\|$ stands for the operator norm; see section 2.3.

Proof of Theorem 3.13. This is a local property. We operate in coordinates and apply Proposition 3.16 (point 1) with $\mathsf{d}=2n$, O^{d} a neighborhood of a point where $p\neq 0$, $X=(x,p)\in\mathbb{R}^{2n}$, and $\mathsf{V}(\theta,X)=\langle p,\mathcal{G}(\theta,x)\rangle$. The rank condition implies that derivatives of all orders of the map $\theta\mapsto\mathsf{V}(\theta,X)$ never vanish at the same point, so that its zeros are isolated and the set $\{\theta\in S^1\ ,\ \mathsf{V}(\theta,X)=0\}$ is finite and a fortiori has measure zero; hence H is continuously differentiable outside $\{p=0\}$. Since $0\le H(x,p)\le k\|p\|$ for some local constant k, the derivative of H^2 is zero at all points $(x,0)$, and, since (37) implies that the norm of $\mathrm{d}H(x,p)$ is bounded at neighboring points where $p\neq 0$, the derivative of H^2 at these points tends to zero as $p\to 0$. H^2 is therefore continuously differentiable everywhere. ∎

Proof of Theorem 3.14. Smoothness outside $\widetilde{\mathcal{Z}}$ is obvious from the expression (5) of H; inequality (34) is a consequence of Proposition 3.16 (point 2), applied with $\mathsf{d}=2n$, $X=(x,p)\in\mathbb{R}^{2n}$, $\mathsf{V}(\theta,X)=\langle p,\mathcal{G}(\theta,x)\rangle$, and O^{d} a neighborhood of a point of $\widetilde{\mathcal{Z}}\setminus\{p=0\}$; it is clear that points (i) and (ii) imply the three conditions (38). ∎

3.5. Application to the minimum time problem. Fix two points x_0, x_1 in M and consider the time optimal problem associated to (1) for $\varepsilon > 0$:

$$(40) \quad (\mathcal{P}_\varepsilon),\ \varepsilon > 0: \qquad \left.\begin{array}{l} \dot{x}(t) = \mathcal{G}(t/\varepsilon, x(t))u(t),\ u(t) \in B^m,\ t \in [0,T], \\ x(0) = x_0,\ x(T) = x_1 \end{array}\right\} \ \min T\,,$$

and the time optimal problem associated to the average system:

$$(41) \qquad (\mathcal{P}_0): \qquad \left.\begin{array}{l} \dot{x}(t) \in \mathcal{E}(x(t)),\ t \in [0,T], \\ x(0) = x_0,\ x(T) = x_1 \end{array}\right\} \ \min T\,.$$

Call $T_\varepsilon(x_0, x_1)$ the minimum time for $(\mathcal{P}_\varepsilon)$, $\varepsilon > 0$, and $T_0(x_0, x_1)$ that for $(\mathcal{P}_0)$; when no confusion arises, we write T_ε and T_0.

Let us develop (40)–(41): concerning (40), T_ε is the infimum of the set of T's such that there is an admissible control $u(.)\colon [0,T] \to B^m$, and $x(.)\colon [0,T] \to M$ satisfying $x(0) = x_0$, $x(T) = x_1$, and $\dot{x}(t) = \mathcal{G}(t/\varepsilon, x(t))u(t)$ for almost all t; Proposition 3.11, point 2 implies that this set is nonempty, and hence T_ε is finite. Concerning (41), T_0 is the infimum of the set of T's such that there is $x(.)\colon [0,T] \to M$ satisfying $x(0) = x_0$, $x(T) = x_1$, and $\dot{x}(t) \in \mathcal{E}(x(t))$ for almost all t. T_0 is finite from Proposition 3.12, point 2. A *solution* to $(\mathcal{P}_\varepsilon)$ (resp., to $(\mathcal{P}_0)$) is $x(.), u(.)$ (resp., $x(.)$) as above with $T = T_\varepsilon$ (resp., $T = T_0$). In general, the minimum T_ε or T_0 need not be reached, i.e., there need not be a solution.

LEMMA 3.17. *Assume the rank condition* (30).

1. *There is a neighborhood $\mathcal{W}$ of any x_1 and two constants $\alpha_0 > 0$ and $C_3 > 0$ such that, for all y in $\mathcal{W}$, $T_\varepsilon(y, x_1) \le 2\pi\varepsilon + C_3 d(x_1, y)$.*
2. *For any x_0, x_1', x_1 in M, one has $T_\varepsilon(x_0, x_1) \le T_\varepsilon(x_0, x_1') + T_\varepsilon(x_1', x_1) + 2\pi\varepsilon$.*

Proof. Apply Proposition 3.11, point 1 with $T = 2\pi$, using as a distance in $\mathcal{W}$ the Euclidean norm in some coordinates: for any two points y, y' in $\mathcal{W}$ such that $\|y - y'\| \le \alpha_0$, there is a control defined on $[0, 2\pi]$, with L^∞ norm smaller than $c_3\|y - y'\|$ that brings y' at time 0 to y at time 2π for system (31); rescaling time and control by ε yields, if $c_3\|y - y'\| \le \varepsilon$, a control with L^∞ norm less than 1 that brings y' at time 0 to y at time $2\pi\varepsilon$ for system (1), and hence, by concatenating controls and using periodicity of $\mathcal{G}$, for any positive integer k, a control with L^∞ norm less than 1 that brings y' at time 0 to y at time $2k\pi\varepsilon$ for system (1) if $c_3\|y - y'\| \le k\varepsilon$. In other words, $T_\varepsilon(y', y) \le 2\pi(\varepsilon + c_3\|y - y'\|)$. Take $y' = x_1$ and $2\pi c_3/C_3$ the ratio between the Euclidean norm and the distance d; this proves point 1. Point 2 follows from using periodicity of $\mathcal{G}$ and concatenating controls while inserting a zero control between time $T_\varepsilon(y', y)$ and the next multiple of 2π. ∎

THEOREM 3.18 (limit of minimum time). *Assume the rank condition* (30).

1. *T_ε is bounded as $\varepsilon \to 0$ and $\limsup_{\varepsilon\to 0} T_\varepsilon \le T_0$.*
2. *If, for $\varepsilon > 0$ small enough, each $(\mathcal{P}_\varepsilon)$ has a solution $x_\varepsilon : [0, T_\varepsilon] \to M$ and there exists a compact $\mathbb{K} \subset M$ such that $x_\varepsilon([0, T_\varepsilon]) \subset \mathbb{K}$ for all $\varepsilon > 0$ small enough, then all accumulation points of the compact family $(x_\varepsilon(.))_{\varepsilon>0}$ in $C^0([0, T_0], M)$ are solutions of $(\mathcal{P}_0)$ and $\lim_{\varepsilon\to 0} T_\varepsilon = T_0$.*

Proof. Consider a minimizing sequence for problem $(\mathcal{P}_0)$, i.e., solutions x^k : $[0, T_0 + \beta_k] \to M$ of the average system (4) with (β_k) a sequence of positive numbers that tends to zero and $x^k(0) = x_0$, $x^k(T_0 + \beta_k) = x_1$ for all k. For each x^k, there is, according to Theorem 3.7, a family $(\,x_\varepsilon^k(.)\,)_{\varepsilon>0}$ such that each $x_\varepsilon^k(.)$ is a solution of (1) with $x_\varepsilon^k(0) = x_0$ and $d(x_\varepsilon^k(t), x^k(t)) \le c_1\varepsilon$ for all t in $[0, T_0 + \beta_k]$. In particular $d(x_\varepsilon^k(T_0 + \beta_k), x^k(t)_1) \le c_1\varepsilon$. Now, from Lemma 3.17, $T_\varepsilon(\,x_\varepsilon^k(T_0 + \beta_k)\,, x_1) \le (2\pi + c_1C_3)\varepsilon$; hence, from the second point of that lemma (with $x_1' = x_\varepsilon^k(T_0 + \beta_k)$), one has $T_\varepsilon = T_\varepsilon(x_0, x_1) \le T_0 + \beta_k + (4\pi + c_1 C_3)\varepsilon$ and, letting k go to infinity, $T_\varepsilon \le T_0 + (4\pi + c_1 C_3)\varepsilon$; this implies point 1. Let us turn to point 2.

Extend x_ε on $[0, \overline{T}]$, with $\overline{T}$ an upperbound of T_ε, by taking $x_\varepsilon(t) = x_1$ for t in $[T_\varepsilon, \overline{T}]$. Any sequence $(x_{\varepsilon_k}(.))_{k\in\mathbb{N}}$ with $\lim \varepsilon_k = 0$ is compact in $C^0([0, \overline{T}], M)$: take a convergent subsequence such that T_{ε_k} also converges to some T^*. The uniform limit goes through x_0 at time 0 and x_1 at time T^* and is, by Theorem 3.7, a solution of the average system (4); hence $T^* \ge T_0$ by definition of T_0. This, together with point 1, implies point 2 because T^* can be any accumulation point of (T_ε) as $\varepsilon \to 0$. ∎

Let us now write the Pontryagin maximum principle [22] for both $(\mathcal{P}_\varepsilon)$, $\varepsilon > 0$, and $(\mathcal{P}_0)$ and see how they are related.

An *extremal* of problem $(\mathcal{P}_\varepsilon)$, $\varepsilon > 0$, is an absolutely continuous map $t \mapsto (x(t), p(t))$ solution to

$$\dot{p} = -\frac{\partial H_\varepsilon}{\partial x}, \quad \dot{x} = \frac{\partial H_\varepsilon}{\partial p} \qquad \text{with} \qquad H_\varepsilon(t, p, x) = \|\langle p, \mathcal{G}(t/\varepsilon, x)\rangle\|. \tag{42}$$

The right-hand side is discontinuous on $\mathcal{S}_\varepsilon = \{(x, p, t),\ \langle p, \mathcal{G}(t/\varepsilon, x)\rangle = 0\}$ (the "switching surface"), where it is in fact not defined.

An *extremal* of $(\mathcal{P}_0)$ is an absolutely continuous $t \mapsto (x(t), p(t))$ solution to

$$\dot{p} = -\frac{\partial H}{\partial x}, \quad \dot{x} = \frac{\partial H}{\partial p}, \tag{43}$$

with H given by (5). The right-hand side is continuous according to Theorem 3.13.

THEOREM 3.19. *If an absolutely continuous map $t \mapsto \overline{x}(t)$ defined on $[0, \overline{T}]$ is a solution of $(\mathcal{P}_\varepsilon)$, $\varepsilon > 0$, (resp., of $(\mathcal{P}_0)$), then there exists $t \mapsto \overline{p}(t)$ defined on $[0, \overline{T}]$ such that $t \mapsto (\overline{p}(t), \overline{x}(t))$ is an extremal of $(\mathcal{P}_\varepsilon)$, $\varepsilon > 0$ (resp., of $(\mathcal{P}_0)$).*

Proof. Problem $(\mathcal{P}_\varepsilon)$, $\varepsilon > 0$, deals with a classical smooth control system; according to [22, 1], the pseudo-Hamiltonian is $h(t, x, p, u) = \langle p, \mathcal{G}(t/\varepsilon, x)\, u\rangle$; an extremal is a curve on the cotangent bundle solution, in local coordinates, of

$$\begin{aligned} \dot{p} &= -\frac{\partial h}{\partial x}(t, x, p, u^*) \ = \ -\left\langle p, \frac{\partial \mathcal{G}}{\partial x} u^* \right\rangle, \\ \dot{x} &= \frac{\partial h}{\partial p}(t, x, p, u^*) \ = \ \mathcal{G}\, u^*, \end{aligned} \tag{44}$$

with $u^*(t)$ a control that maximizes the pseudo-Hamiltonian for almost all time; it is defined by $u^* = \frac{\langle p, \mathcal{G}\rangle}{\|\langle p, \mathcal{G}\rangle\|}$ if $\langle p, \mathcal{G}(t/\varepsilon, x)\rangle \ne 0$; the maximized Hamiltonian $H_\varepsilon(t, p, x) = \max_u h(t, x, p, u)$ is that in (42), and (44) is then the differential equation (42), whose right-hand side is discontinuous at points where $\langle p, \mathcal{G}(t/\varepsilon, x)\rangle$ vanishes.

Let us now turn to $(\mathcal{P}_0)$. Since the set of admissible velocities is not a priori smooth with respect to the state variable, we use a nonsmooth version of the Pontryagin maximum principle for differential inclusions that we recall for self-containedness.

> Theorem 9.1 in [11, Chapter 4]: *If $\dot{x} \in \mathcal{E}(x)$ is a locally Lipschitz differential inclusion and $t \mapsto \overline{x}(t)$ is an absolutely continuous function defined on $[0, \overline{T}]$ solution to the problem (41), then there exists $t \mapsto \overline{p}(t)$ defined on $[0, \overline{T}]$ such that $(-\dot{\overline{p}}, \dot{\overline{x}}) \in \partial_C H(\overline{x}, \overline{p})$ for almost all $t \in [0, \overline{T}]$ with $H(x, p) = \max_{v\in\mathcal{E}(x)} \langle p, v\rangle$ and $\partial_C H$ the generalized gradient of H.*

The set-valued map $\mathcal{E}(.)$ in (3) is indeed locally Lipschitz: in local coordinates, for x_1, x_2 in $\mathbb{R}^n$, denoting by δ the Hausdorff distance between two sets, one has

$$\begin{aligned}\delta\left(\mathcal{E}(x_1), \mathcal{E}(x_2)\right) &= \max\left\{\sup_{v_1\in\mathcal{E}(x_1)}\inf_{v_2\in\mathcal{E}(x_2)}\|v_1-v_2\|, \sup_{v_2\in\mathcal{E}(x_2)}\inf_{v_1\in\mathcal{E}(x_1)}\|v_1-v_2\|\right\}\\ &= \max_{\|\mathcal{U}_1\|_\infty\le 1}\min_{\|\mathcal{U}_2\|_\infty\le 1}\|\overline{\mathcal{G}}(x_1,\mathcal{U}_1)-\overline{\mathcal{G}}(x_2,\mathcal{U}_2)\| \;\le\; \operatorname{Lip}\mathcal{G}\;\|x_1-x_2\|.\end{aligned}$$

According to (8), the Hamiltonian H defined in the above quoted theorem coincides with the map H defined in (5). □

Remark 3.20. This result and Theorem 3.18 have two interpretations:

1. They prove that the operations of *averaging* and *computing the Hamiltonian for the minimum time problem* commute. Indeed, the Hamiltonian H was obtained by applying the maximum principle to problem (41), i.e., minimum time for the average system (4), but it is also the average of that in (42) with respect to the fast variable.

2. They prove indirectly an averaging result for the minimum time control problem (41); the averaging techniques in [10] do not apply to minimum time because they require smoothness of the Hamiltonian, while averaging is used in [14, 13] for minimum time with only partial theoretical justifications but numerical evidence of efficiency.

Let us now focus on the differential equations (43) that govern the extremals of $(\mathcal{P}_0)$. It is of great importance to know whether these equations define a Hamiltonian flow on T^*M, i.e., whether solutions through all initial conditions are unique or not. The right-hand side of (43) is continuous because, from Theorem 3.13, H is continuously differentiable; this ensures existence of solutions. We saw that H is smooth (C^∞) on $\mathrm{T}^*M \setminus \widetilde{\mathcal{Z}}$ (see (33)); hence solutions through points outside $\widetilde{\mathcal{Z}}$ are always unique. The following result gives uniqueness of solutions even on $\widetilde{\mathcal{Z}}$ in the least degenerated case possible.

THEOREM 3.21 (Hamiltonian flow for $(\mathcal{P}_0)$). *Assume that the rank condition* (30) *holds, as well as conditions* (i) *and* (ii) *in Theorem* 3.14. *Then the differential equation* (43) *has a unique solution from any initial condition.*

Proof. For an autonomous ODE $\dot z = f(z)$ in a finite dimensional space, where f satisfies $\|f(z_1)-f(z_2)\| \le \omega(\|z_1-z_2\|)$ with $\omega : [0,+\infty) \to [0,+\infty)$ nondecreasing, Kamke uniqueness theorem [15, Chap. III, Thm. 6.1] states that uniqueness of solutions holds if $\int_0^\alpha \frac{\mathrm{d}u}{\omega(u)} = +\infty$ for arbitrarily small $\alpha > 0$. From Theorem 3.14, we are in this case with $\omega(u) = c\,u\ln(1/u)$, and $\int \frac{\mathrm{d}u}{\omega(u)} = -\frac{1}{c}\,\ln\ln(\frac{1}{u})$. □

Proving existence of a flow for (43) in more general situations (weaker sufficient condition) than this theorem is an interesting program to be pursued. However, it turns out to be applicable to the control of orbit transfer with low thrust; see section 5.

Point (ii) is very mild and states only that the control vector fields are linearly independent. Point (i) is more artificial: the fact that $\langle p, \mathcal{G}(\theta,x)\rangle = 0$ has at most one solution θ has to be checked by hand, while the fact that $\partial\mathcal{G}/\partial\theta$ does not vanish at the same time is equivalent to the rank condition

$$\operatorname{rank}\left\{\mathcal{G}(\theta,x), \frac{\partial\mathcal{G}}{\partial\theta}(\theta,x)\right\} = \dim\Big(\operatorname{Range}\mathcal{G}(\theta,x) + \operatorname{Range}\frac{\partial\mathcal{G}}{\partial\theta}(\theta,x)\Big) = n. \tag{45}$$

It is true for the Kepler problem and is used in [9] to show that the discontinuities in (42) are always "π-singularities"; i.e., the control u^* switches to its opposite.

4. Kepler control systems. We call a *Kepler control system* with small control a family of control systems on $S^1 \times M$ of the form

$$(\mathcal{K}_\varepsilon) \quad \left\{ \begin{array}{rcl} \dot\theta & = & \omega(\theta,x) + g(\theta,x)\,v, \\ \dot x & = & G(\theta,x)\,v, \end{array} \right. \qquad \|v\| \le \varepsilon, \tag{46}$$

where G and g can be viewed, with the same convention as in (1), as $n\times m$ and $1\times m$ matrices smoothly depending on (θ,x), and ω is a smooth function $S^1\times M \to \mathbb{R}$ that remains larger than a strictly positive constant:

$$\omega(\theta,x) \ge k_\omega > 0 \quad \text{ for all } (\theta,x)\in S^1\times M. \tag{47}$$

In fact, this is an affine control system on $S^1\times M$,

$$\dot\xi = f_0(\xi) + \sum_{i=1}^m v_i f_i(\xi), \tag{48}$$

with $\xi=(\theta,x)$, $f_0 = \omega\,\frac{\partial}{\partial\theta}$, and, for $1\le i\le m$, the smooth vector field f_i is represented by the ith column of the matrix notation G and g. If, in (48), one assumes only that all solutions of $\dot\xi = f_0(\xi)$ are periodic, additional conditions are needed for the orbits to induce a nice foliation that splits the state manifold into a product $M\times S^1$.

4.1. Relation with fast-oscillating systems. For a solution $t\mapsto(\theta(t),x(t))$ of $(\mathcal{K}_\varepsilon)$ in (46), let $\Theta(t)$ be the cumulated angle, i.e., $\Theta(.)$ is continuous $[0,T]\to\mathbb{R}$ with $\Theta(t)\equiv\theta(t) \bmod 2\pi$ for all t and $\Theta(0)\in[0,2\pi)$, and define a new "time"

$$\lambda = \mathcal{R}(t) \overset{\Delta}{=} \varepsilon\left(\Theta(t)-\Theta(0)\right). \tag{49}$$

Taking ε_0 small enough so that $|\omega(\theta,x)+\varepsilon\, g(\theta,x)\,u| > k_\omega/2$ for $x\in\mathbb{K}$, $\|u\|\le 1$, $\varepsilon<\varepsilon_0$, one has $\mathrm{d}\mathcal{R}/\mathrm{d}t > \varepsilon\, k_\omega/2$, and hence $\mathcal{R}$ is strictly increasing and one-to-one, and

$$\frac{k_\omega}{2}\varepsilon\, t \le \mathcal{R}(t) \le k_\omega\,\varepsilon\, t \quad \text{with } k_\omega = \sup_{S^1\times\mathbb{K}} \omega + \varepsilon_0 \sup_{S^1\times\mathbb{K}} \|g\|\,. \tag{50}$$

Then $\lambda\mapsto\widetilde{x}(\lambda)=x(\mathcal{R}^{-1}(\lambda))$ is a solution of the system

$$(\widetilde\Sigma_{\theta_0,\varepsilon}) \qquad \frac{\mathrm{d}\widetilde x}{\mathrm{d}\lambda} = \frac{G(\theta_0+\frac{\lambda}{\varepsilon},\widetilde x)\,\widehat u}{\omega(\theta_0+\frac{\lambda}{\varepsilon},\widetilde x) \;+\; \varepsilon\, g(\theta_0+\frac{\lambda}{\varepsilon},\widetilde x)\,\widehat u}, \qquad \|\widehat u\|\le 1, \tag{51}$$

associated with the control $\lambda\mapsto\widehat u(\lambda) = v(\mathcal{R}^{-1}(\lambda))/\varepsilon$. Except for the term $\varepsilon g\widehat u$ in the denominator, this is a fast-oscillating system (1) with $\mathcal{G}=G/\omega$. We now apply section 3.

4.2. Average control system. The definition uses $\overline\omega$ defined by

$$\frac{1}{\overline\omega(x)} = \frac{1}{2\pi}\int_0^{2\pi} \frac{\mathrm{d}\theta}{\omega(\theta,x)}\ . \tag{52}$$

DEFINITION 4.1 (average control system of Kepler control systems). *The average control system of the Kepler control system* (46) *is the differential inclusion*

$$\dot x \in \mathcal{E}(x) \tag{53}$$

with $\mathcal{E}$ defined by (3) *using* $\overline{\mathcal{G}} : M \times L^\infty(S^1, \mathbb{R}^m) \to \mathrm{T}M$ *defined by*

$$\overline{\mathcal{G}}(x, \mathcal{U}) \;=\; \overline{\omega}(x)\, \frac{1}{2\pi}\int_0^{2\pi} \frac{G(\theta, x)}{\omega(\theta, x)}\, \mathcal{U}(\theta)\mathrm{d}\theta \tag{54}$$

instead of (2)*. Solutions are defined as in Definition* 3.2*.*

Remark 4.2. This is almost Definition 3.2 applied to (51), which is equivalent to (46) via time changes, *except*

(i) the term $\varepsilon g\widehat{u}$ in the denominator of (51) has been discarded,

(ii) the right-hand side has been multiplied by $\overline{\omega}(x)$.

4.3. Convergence theorem. The counterpart of Theorem 3.7 is the following.

THEOREM 4.3 (convergence for Kepler control systems).

1. *Let* $x_0(.) : [0, T] \to M$ *be an arbitrary solution of* (53)*, and let* $\theta^0 \in S^1$*. There exist a family of measurable functions* $\overline{u}_\varepsilon(.) : [0, T] \to B^m$*, indexed by* $\varepsilon > 0$*, and positive constants* c, ε_0*, such that, if* $t \mapsto (\theta_\varepsilon(t), x_\varepsilon(t))$ *is the solution of* (46) *with control* $u = \overline{u}_\varepsilon(t)$ *and initial condition* $(\theta_\varepsilon(0), x_\varepsilon(0)) = (\theta^0, x_0(0))$*, it is defined on* $[0, T/\varepsilon]$ *for* ε *smaller than* ε_0 *and*

$$d(\, x_\varepsilon(t)\, ,\, x_0(\varepsilon t)) < c\,\varepsilon\,, \quad t \in \left[0, \frac{T}{\varepsilon}\right], \quad 0 < \varepsilon < \varepsilon_0\,, \tag{55}$$

and thus $\tau \mapsto x_\varepsilon(\tau/\varepsilon)$ *converges uniformly on* $[0, T]$ *to* $\tau \mapsto x_0(\tau)$ *when* ε *tends to zero.*

2. *Let* $\mathbb{K}$ *be a compact subset of* M*,* $(\varepsilon_n)_{n\in\mathbb{N}}$ *a decreasing sequence of positive real numbers converging to zero, and* $\big(\theta_n(.), x_n(.)\big) : [0, T/\varepsilon_n] \to S^1 \times \mathbb{K}$ *a solution of system* (46) *for each* n*, with* $\varepsilon = \varepsilon_n$ *and some control* $u = u_n(t)$*,* $u_n(.) \in L^\infty([0, T/\varepsilon_n], \mathbb{R}^m)$*,* $\|u_n(.)\|_\infty \le 1$*. Then the sequence* $\big(\tau \mapsto (x_n(\tau/\varepsilon_n))\big)_{n\in\mathbb{N}}$ *is compact for the topology of uniform convergence on* $[0, T]$*, and the limit of any converging subsequence is a solution* $x^*(.)$ *of the average differential inclusion* (53)*.*

Proof. We assume that M is $\mathbb{R}^n$, d is the Euclidean distance, and all vector fields have a common compact support, and hence all maps share a global Lipschitz constant and a global bound; by "a constant" we mean a number that depends only on these bounds and Lipschitz constants. It is left to the reader to check that, as for the proof of Theorem 3.7, the present proof extends to M with any distance d described in section 2.3.

Let $\tau \mapsto x_0(\tau)$ be a solution of (53) on $[0, T]$. Define $\mathcal{P}(.)$ by

$$\mathcal{P}(\tau) = \int_0^\tau \overline{\omega}\,(x_0(\mathrm{t}))\,\mathrm{dt} \tag{56}$$

and $\widehat{x}_0(.)$ by $\widehat{x}_0(\lambda) = x_0(\mathcal{P}^{-1}(\lambda))$. The latter is a solution on $[0, \mathcal{P}(T)]$ of

$$\frac{\mathrm{d}\widehat{x}_0}{\mathrm{d}\lambda} \in \frac{1}{\overline{\omega}(\widehat{x}_0)}\mathcal{E}(\widehat{x}_0) \tag{57}$$

with $\mathcal{E}$ defined by (3) and (54). This is the average system (in the sense of Definition 3.2) of the fast-oscillating control system

$$(\widehat{\Sigma}_{\theta_0,\varepsilon}) \qquad \frac{\mathrm{d}\widehat{x}}{\mathrm{d}\lambda} = \frac{G(\theta_0 + \frac{\lambda}{\varepsilon}, \widehat{x})\, \widehat{u}}{\omega(\theta_0 + \frac{\lambda}{\varepsilon}, \widehat{x})}\;, \quad \|\widehat{u}\| \le 1\;. \tag{58}$$

Theorem 3.7 (point 1) yields a family of controls $\widehat{u}_\varepsilon$ such that the solutions $\widehat{x}_\varepsilon(.)$ of $(\widehat{\Sigma}_{\theta_0,\varepsilon})$ with initial condition $\widehat{x}_0(0)$ and control $\widehat{u}_\varepsilon$ converge to $\widehat{x}_0(.)$ uniformly:

$$d(\widehat{x}_\varepsilon(\lambda),\widehat{x}_0(\lambda)) \leq c'\,\varepsilon \quad \text{for all } \lambda\in[0,\mathcal{P}(T)] \tag{59}$$

for some constant c'. For each ε, let $\widetilde{x}_\varepsilon(.)$ be the solution of $(\widetilde{\Sigma}_{\theta_0,\varepsilon})$—see (51)—with the same initial condition and the same control. Since (51) can be rewritten as

$$\frac{\mathrm{d}\widetilde{x}}{\mathrm{d}\lambda} = \Big(1-\frac{\varepsilon\, g(\theta_0+\frac{\lambda}{\varepsilon},\widetilde{x})\,\widehat{u}}{\omega(\theta_0+\frac{\lambda}{\varepsilon},\widetilde{x})+\varepsilon\, g(\theta_0+\frac{\lambda}{\varepsilon},\widetilde{x})\,\widehat{u}}\Big)\frac{G(\theta_0+\frac{\lambda}{\varepsilon},\widetilde{x})\,\widehat{u}}{\omega(\theta_0+\frac{\lambda}{\varepsilon},\widetilde{x})}\,, \tag{60}$$

the norm of the difference between the right-hand sides of $(\widetilde{\Sigma}_{\theta_0,\varepsilon})$ and $(\widehat{\Sigma}_{\theta_0,\varepsilon})$ is bounded by $k\,\varepsilon$ for some constant $k>0$; classical theorems on smooth dependence of solutions on "parameters" yield some constant c'' such that

$$d(\widetilde{x}_\varepsilon(\lambda),\widehat{x}_\varepsilon(\lambda)) \leq c''\,\varepsilon \quad \text{for all } \lambda\in[0,\mathcal{P}(T)]\,. \tag{61}$$

Then define

$$t=\mathfrak{T}(\lambda)=\frac{1}{\varepsilon}\int_0^\lambda \frac{\mathrm{d}\ell}{\omega(\theta_0+\frac{\ell}{\varepsilon},\widetilde{x}_\varepsilon(\ell))+\varepsilon\, g(\theta_0+\frac{\ell}{\varepsilon},\widetilde{x}_\varepsilon(\ell))\,\widehat{u}_\varepsilon(\ell)} \tag{62}$$

and the controls $t\mapsto\overline{u}_\varepsilon(t)$ by $\widehat{u}_\varepsilon(\lambda)=\overline{u}_\varepsilon(\mathfrak{T}(\lambda))$; the solutions $x_\varepsilon(.)$ of (46) with these controls are given by $\widetilde{x}_\varepsilon(\lambda)=x_\varepsilon(\mathfrak{T}(\lambda))$, and one therefore has

$$d(x_\varepsilon(\mathfrak{T}\circ\mathcal{P}(\tau)),x_0(\tau)) < (c'+c'')\varepsilon,\quad \tau\in[0,T]\,. \tag{63}$$

Now, on the one hand, (62) yields

$$\begin{aligned}\mathfrak{T}(\mathcal{P}(\tau)) &= \frac{1}{\varepsilon}\int_0^{\mathcal{P}(\tau)}\frac{\mathrm{d}\ell}{\omega(\theta_0+\frac{\ell}{\varepsilon},\widehat{x}_\varepsilon(\ell))}+\rho\\ \text{with}\quad \rho &= \frac{1}{\varepsilon}\int_0^{\mathcal{P}(\tau)}\left(\frac{\widehat{\omega}(\ell)-\widetilde{\omega}(\ell)}{\widehat{\omega}(\ell)}+\frac{\varepsilon\, g(\ell)\,\widehat{u}_\varepsilon(\ell)}{\widetilde{\omega}(\ell)+\varepsilon\, g(\ell)\,\widehat{u}_\varepsilon(\ell)}\right)\frac{\mathrm{d}\ell}{\widetilde{\omega}(\ell)},\end{aligned} \tag{64}$$

where $\widetilde{\omega}(\ell)$ stands for $\omega(\theta_0+\frac{\ell}{\varepsilon},\widetilde{x}_\varepsilon(\ell))$, $\widehat{\omega}(\ell)$ stands for $\omega(\theta_0+\frac{\ell}{\varepsilon},\widehat{x}_\varepsilon(\ell))$, and $g(\ell)$ stands for $g(\theta_0+\frac{\ell}{\varepsilon},\widetilde{x}_\varepsilon(\ell))$; using (61) and Lipschitz continuity of ω to bound the first term in the integral, this implies that $|\rho|$ is bounded by a constant. On the other hand, one has, according to (56), $\tau=\int_0^{\mathcal{P}(\tau)}\mathrm{d}\lambda/\overline{\omega}(\widehat{x}_0(\lambda))$. Developing $\overline{\omega}$ according to its definition (52), in which we add $\theta_0+\frac{\lambda}{\varepsilon}$ to θ without changing the integral due to periodicity, τ is also equal to $\frac{1}{2\pi}\iint_{\theta\in S^1,\,0\leq\lambda\leq\mathcal{P}(\tau)}\mathrm{d}\lambda\,\mathrm{d}\theta/\omega(\theta_0+\frac{\lambda}{\varepsilon}+\theta,\widehat{x}_0(\lambda))$. Finally, performing the change of variable $\lambda=\ell-\varepsilon\mu(\theta)$, with $\mu(\theta)$ defined in section 2.4, yields

$$\tau=\int_{\varepsilon\mu(\theta)}^{\mathcal{P}(\tau)+\varepsilon\mu(\theta)}\left(\frac{1}{2\pi}\int_0^{2\pi}\frac{\mathrm{d}\theta}{\omega(\theta_0+\frac{\ell}{\varepsilon},\widehat{x}_0(\ell-\varepsilon\mu(\theta)))}\right)\mathrm{d}\ell.$$

Using (59), the fact that $|\mu(\theta)|<2\pi$, and Lipschitz continuity of both $\widehat{x}_0$ and ω, we deduce from this and (64) that $\left|\mathfrak{T}\circ\mathcal{P}(\tau)-\frac{\tau}{\varepsilon}\right|\leq|\rho|+k'$ for some constant k', and finally, using the fact that $x_\varepsilon(.)$ is Lipschitz continuous with constant $2\,\varepsilon\sup\|G\|/k_\omega$, one has $d(x_\varepsilon(\mathfrak{T}\circ\mathcal{P}(\tau)),x_\varepsilon(\frac{\tau}{\varepsilon}))<c'''\varepsilon$ for some constant c'''. This and (63) imply point 1 of the theorem, with $c=c'+c''+c'''$ in (55).

For point 2, consider $\big(\theta_n(.), x_n(.)\big) : [0, T/\varepsilon_n] \to S^1 \times \mathbb{K}$ a solution of system (46) with $\varepsilon = \varepsilon_n$ and some control $u = u_n(t)$. Following (49)–(51) and setting $\lambda = \mathcal{R}_n(t)$ (we write $\mathcal{R}_n$ because $\mathcal{R}$ in (49) is constructed for system $(\widetilde{\Sigma}_{\theta_0,\varepsilon_n})$ and thus depends on n), one associates to these x and u a control $\lambda \mapsto \widetilde{u}_n(\lambda)$ and a solution $\lambda \mapsto \widetilde{x}_n(\lambda)$ of $(\widetilde{\Sigma}_{\theta_0,\varepsilon_n})$. The solutions $\lambda \mapsto \widehat{x}_n(\lambda)$ of $(\widehat{\Sigma}_{\theta_0,\varepsilon_n})$ with the same control and initial condition satisfy, for the same reasons as (61), $d(\widehat{x}_n(\lambda), \widetilde{x}_n(\lambda)) < c''\varepsilon_n$ for some constant c''. By Theorem 3.7 (point 2), the sequence $(\widehat{x}_n)$ is compact, and subsequences converge to solutions $\lambda \mapsto \widehat{x}_0(\lambda)$ of (57), and hence the same subsequences of $(\widetilde{x}_n)$ converge as well, and, with $\tau = \mathcal{Q}(\lambda) \triangleq \int_0^\lambda \frac{\mathrm{d}\ell}{\overline{\omega}(\widehat{x}(\ell))}$, the maps $\tau \mapsto \widetilde{x}_n(\mathcal{Q}^{-1}(\tau)) = x_n((\mathcal{Q} \circ \mathcal{R}_n)^{-1}(\tau))$ converge to a solution $\tau \mapsto x_0(\tau) = \widehat{x}_0(\mathcal{Q}^{-1}(\tau))$ of the average system (53), with distance less than $c'\varepsilon_n$ for some constant c'. Using the same argument as in point 1 for $\mathfrak{T} \circ \mathcal{P}(\tau)$, one gets a bound for $|(\mathcal{Q} \circ \mathcal{R}_n)^{-1}(\tau) - \frac{\tau}{\varepsilon_n}|$ and, for some constant c''', $d(\, x_n((\mathcal{Q} \circ \mathcal{R}_n)^{-1}(\tau))\, ,\, x_n(\frac{\tau}{\varepsilon})) \le c'''\varepsilon_n$. Point 2 is proved. ☐

4.4. Dimension of $\mathcal{E}(x)$. In section 3.3, and in particular in Proposition 3.10, $\mathcal{G}$ can simply be replaced with G. However, it is interesting to give a more intrinsic characterization of $r(\theta, x)$ and thus of $\dim \mathcal{E}(x)$.

PROPOSITION 4.4. *If* (46) *and* (48) *represent the same control system, then*

$$\dim\Bigg(\sum_{j\in\mathbb{N}} \operatorname{Range} \frac{\partial^j G}{\partial\theta^j}(\theta, x)\Bigg)$$
$$= -1 + \operatorname{rank}\Big(\{f_0(\theta, x)\} \cup \Big\{\operatorname{ad}^j_{f_0} f_k\,(\theta, x),\ j \in \mathbb{N}, 1 \le k \le m\Big\}\Big). \tag{65}$$

Proof. Straightforward computation using the fact that $f_0 = \partial/\partial\theta$. ☐

Note that the right-hand side is $r(\theta, x)$. Proposition 3.10 applies, with this definition of r. In particular, the "full rank case" becomes the following.

PROPOSITION 4.5. *If the vector fields f_0 and $\operatorname{ad}^j_{f_0} f_k$, $1 \le k \le m$, $j \in \mathbb{N}$, span the whole tangent space of $S^1 \times M$, then $\mathcal{E}(x)$ has a nonempty interior for all x.*

4.5. The function $H(x, p)$. Instead of (5), H has to be taken as follows, with $\overline{\omega}$ defined as in (52):

$$H(x, p) = \overline{\omega}(x)\, \frac{1}{2\pi}\int_0^{2\pi} \left\| \left\langle p, \frac{G(\theta, x)}{\omega(\theta, x)} \right\rangle \right\| \mathrm{d}\theta\,. \tag{66}$$

The characterization of $\mathcal{E}(x)$ in Proposition 3.4 is unchanged. In the "full rank case" the results from section 3.4 on the degree of differentiability apply without a change.

4.6. Application to the minimum time problem. As in section 3.5, but now for the Kepler system (46), let x_0, x_1 be fixed, call T_ε the minimum time such that, from *some* θ_0, θ_1, (θ_1, x_1) can be reached from (θ_0, x_0) in system $(\mathcal{K}_\varepsilon)$ (obviously $T_\varepsilon \to +\infty$ as $\varepsilon \to 0$), and call T_0 the minimum time such that x_1 can be reached from x_0 in the average system (53). The equivalent of Theorem 3.18, with a similar proof, using Theorem 4.3, is as follows.

THEOREM 4.6. *In the full rank case, one has $\limsup_{\varepsilon\to 0} \varepsilon T_\varepsilon \le T_0$ (hence $\varepsilon\, T_\varepsilon$ is bounded as $\varepsilon \to 0$). If, for all $\varepsilon > 0$ small enough, there is a minimizing solution $(\theta_\varepsilon, x_\varepsilon) : [0, T_\varepsilon] \to S^1 \times M$ and they all remain in a common compact subset of M, then all accumulation points (as $\varepsilon \to 0$) of the compact family $(\tau \to x_\varepsilon(\frac{\tau}{\varepsilon}))_{\varepsilon>0}$ in $C^0([0, T_0], M)$ are minimizing for the average system and $\lim_{\varepsilon\to 0} \varepsilon\, T_\varepsilon = T_0$.*

The Hamiltonian for minimum time for the average system is given by (66); one has to perform the time scaling described in section 4.1 to have a result like Theorem 3.21 and the simple "commutation between averaging and writing Hamiltonian" noted in Remark 3.20. Let us translate in terms of (46) the sufficient condition for existence of a Hamiltonian flow given by Theorem 3.21.

THEOREM 4.7. *In the full rank case, assume that* $\langle p, G(\theta,x)\rangle$ *and* $\langle p, \partial G/\partial\theta(\theta,x)\rangle$ *do not vanish simultaneously outside* $\{p=0\}$, *that* $\theta\mapsto\langle p, G(\theta,x)\rangle$ *vanishes at most once for each* $(x,p)\in \mathrm{T}^*M$, $p\neq 0$, *and that* $\operatorname{rank}\mathcal{G}(\theta,x)=m$ *for each* $(\theta,x)\in S^1\times M$. *Then* (43), *with* H *given by* (66), *has a unique solution for any initial condition.*

The discussion that follows Theorem 3.21 also applies to the above; let us mention that, once it has been checked that, for each (x,p), $\langle p, G(\theta,x)\rangle$ vanishes for at most one θ, the other conditions are guaranteed if (45) holds with $\mathcal{G}$ replaced by G or, in terms of the vector fields in (48), if, for all $\xi=(\theta,x)$,

$$\operatorname{rank}\{f_0(\xi), f_1(\xi),\dots,f_m(\xi), \mathrm{ad}_{f_0}f_1(\xi),\dots,\mathrm{ad}_{f_0}f_m(\xi)\}=n+1. \tag{67}$$

We prove in the next section that the above conditions are true for the planar control 2-body problem.

5. Application to the controlled 2-body system.

In this section we study some properties of the planar control system and demonstrate that it satisfies the condition of Theorem 3.21 on the domain of nondegenerated elliptic orbits.

5.1. Planar control 2-body system.

It is classically described by some first integrals of the free movement—here the semimajor axis a and the eccentricity vector (e_x, e_y)—and one angle L following the dynamics; we restrict our attention to the set of nondegenerated elliptic orbits rotating in the direct sense; i.e., the state space is $S^1\times M$ with $M=\{(a,e_x,e_y)\in\mathbb{R}^3,\ a>0 \text{ and } {e_x}^2+{e_y}^2<1\}$. The control $u=(u_t,u_n)$ is expressed in the tangential-normal frame, and the system reads as

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} a\\ e_x\\ e_y\\ L\end{pmatrix}=\frac{1}{a^{3/2}}\begin{pmatrix}0\\0\\0\\ \mathtt{w}(e_x,e_y,L)\end{pmatrix}+\sqrt{a}\begin{pmatrix} 2\,a\,\mathtt{a}_a(e_x,e_y,L) & 0\\ 2\,\mathtt{a}_x(e_x,e_y,L) & \mathtt{b}_x(e_x,e_y,L)\\ 2\,\mathtt{a}_y(e_x,e_y,L) & \mathtt{b}_y(e_x,e_y,L)\\ 0 & 0\end{pmatrix}\begin{pmatrix}u_t\\u_n\end{pmatrix} \tag{68}$$

with

$$\begin{aligned}
\mathtt{w}(e_x,e_y,L)&=\frac{(1+e_x\cos L+e_y\sin L)^2}{(1-e^2)^{3/2}},\\
\mathtt{a}_a(e_x,e_y,L)&=\frac{\sqrt{1+e^2+2\,e_x\cos L+2\,e_y\sin L}}{\sqrt{1-e^2}},\\
\mathtt{a}_x(e_x,e_y,L)&=\frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2\,e_x\cos L+2\,e_y\sin L}}(e_x+\cos L),\\
\mathtt{a}_y(e_x,e_y,L)&=\frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2\,e_x\cos L+2\,e_y\sin L}}(e_x+\cos L),\\
\mathtt{b}_x(e_x,e_y,L)&=\frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2\,e_x\cos L+2\,e_y\sin L}}\\
&\qquad\qquad\times\frac{-2\,e_y+(e_x^2-e_y^2-1)\sin L-2\,e_xe_y\cos L}{1+2\,e_x\cos L+2\,e_y\sin L},
\end{aligned}$$

$$\mathtt{b}_y(e_x,e_y,L)=\frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2\,e_x\cos L+2\,e_y\sin L}}\times\frac{2\,e_x+(e_x^2-e_y^2+1)\cos L+2\,e_xe_y\sin L}{1+2\,e_x\cos L+2\,e_y\sin L}.$$

The eccentricity e is the norm of the eccentricity vector, $e=\sqrt{e_x^2+e_y^2}$. Low thrust translates into $\|u\|\leq\varepsilon$ for a small ε.

Remark 5.1. This is indeed a "Kepler control system" of the type (46) except that $\omega=\mathtt{w}/a^{3/2}$ is, although strictly positive, not bounded from below by a positive constant on $S^1\times M$. There is such a lowerbound if one replaces M by $M^{\bar c}=\{(a,e_x,e_y)\in\mathbb{R}^3,\ a>0,\ \text{and}\, {e_x}^2+{e_y}^2<\bar c\}$ with $\bar c<1$. Strictly speaking, the results of the paper have to be applied in $M^{\bar c}$, $\bar c<1$. However, Theorems 4.3 or 4.7, for instance, may be applied in M because each statement may ultimately be restricted to a compact subset of M, itself included in some $M^{\bar c}$, $\bar c<1$.

The Hamiltonian that both defines the average system according to (6) and yields the Hamiltonian system governing extremals for minimum time is given by (66). Since $\int_0^{2\pi}\mathrm{d}L/\mathtt{w}(e_x,e_y,L)=2\pi$, it can be expressed as $H(a,e_x,e_y,p_a,p_{e_x},p_{e_y})=\sqrt{a}\mathcal{H}(e_x,e_y,ap_a,p_{e_x},p_{e_y})$ with

$$\mathcal{H}(e_x,e_y,A,X,Y)=\frac{1}{2\pi}\int_0^{2\pi}\|(A\,X\,Y)\mathtt{G}(e_x,e_y,L)\|,$$
$$\mathtt{G}(e_x,e_y,L)=\begin{pmatrix}2\,\mathtt{a}_a/\mathtt{w} & 0\\ 2\,\mathtt{a}_x/\mathtt{w} & \mathtt{b}_x/\mathtt{w}\\ 2\,\mathtt{a}_y/\mathtt{w} & \mathtt{b}_y/\mathtt{w}\end{pmatrix}.$$

5.2. Hamiltonian flow. Theorem 4.7 applies to this system. Indeed, we have the following.

PROPOSITION 5.2. *For each (e_x,e_y,a), with ${e_x}^2+{e_y}^2<1$ and $a>0$, and each $(A,X,Y)\neq(0,0,0)$, the vector $(A\,X\,Y)\mathtt{G}(e_x,e_y,L)$ vanishes for at most one angle L.*

Proof. Removing denominators, the equations $A\mathtt{a}_a+X\mathtt{a}_x+Y\mathtt{a}_y=0$ and $X\mathtt{b}_x+Y\mathtt{b}_y=0$ can be written as

$$\begin{aligned}\big(2\,e_xA+2(1-e^2)X\big)\cos L+\big(2\,e_yA+2(1-e^2)Y\big)\sin L&\\ =-(1+e^2)A-2\,e_x(1-e^2)X-2\,e_y(1-e^2)Y,&\\ \big(-2\,e_xe_yX+(e_x^2-e_y^2+1)Y\big)\cos L+\big((e_x^2-e_y^2-1)X+2\,e_xe_yY\big)\sin L&\\ =2\,e_yX-2\,e_xY.&\end{aligned}$$

If

$$\Delta=\begin{vmatrix}2\,e_xA+2(1-e^2)X & 2\,e_yA+2(1-e^2)Y\\ -2\,e_xe_yX+(e_x^2-e_y^2+1)Y & (e_x^2-e_y^2-1)X+2\,e_xe_yY\end{vmatrix}$$

is nonzero, there is clearly at most one solution L. If $\Delta=0$, there exists $\lambda\neq0$ such that

$$\begin{aligned}2\,e_xA+2(1-e^2)X&=\lambda\left(-2\,e_xe_yX+(e_x^2-e_y^2+1)Y\right),\\ 2\,e_yA+2(1-e^2)Y&=\lambda\left((e_x^2-e_y^2-1)X+2\,e_xe_yY\right),\end{aligned}$$

and there may be a solution to the system in $\cos L$ and $\sin L$ only if the right-hand side components are proportional with respect to λ,

$$(1+e^2)A + 2\,e_x(1-e^2)X + 2\,e_y(1-e^2)Y = -2\,\lambda(e_yX - e_xY).$$

These three equations form a linear system in (A,X,Y), $M(A,X,Y)^T = 0$ with

$$M = \begin{pmatrix} 2e_x & 2(1-e^2+\lambda e_xe_y) & -\lambda(e_x^2-e_y^2+1) \\ 2e_y & -\lambda(e_x^2-e_y^2-1) & 2(1-e^2-\lambda e_xe_y) \\ (1+e^2) & 2\left(e_x(1-e^2)+\lambda e_y\right) & 2\left(e_y(1-e^2)-\lambda e_x\right) \end{pmatrix}.$$

A brief computation gives $\det M = (1-e)^3(1+e)^3(\lambda^2+4)$, strictly positive when $0 \le e < 1$. Hence $M(A,X,Y)^T = 0$ implies $(A,X,Y) = 0$. ∎

Since the rank of $\mathtt{G}$ is obviously equal to 2 and the rank of $\{\mathtt{G}, \partial\mathtt{G}/\partial L\}$ equal to 3 for any (e_x,e_y,L), the hypotheses of Theorem 4.7 are satisfied by the planar control 2-body system, and it guarantees existence of a flow for the Hamiltonian system governing the extremals of minimum time for its average system.

6. Conclusion.

Attempting to formulate a control theory equivalent to the averaging theorems for ODEs naturally leads to, and justifies, the notion of an average control system introduced in this paper. It has a conceptual importance as well as, for instance, applications to approximation of minimum time control.

Besides its definition and description, we gave results on its regularity and on the dimension of its velocity set ("number of inputs"). However, these are mostly a starting point. The regularity of H has to be further explored when the conditions of Theorem 3.21 do not hold; see the last paragraph of section 3.5.

It has already allowed us to give (with restrictions on the eccentricities; see Remark 5.1) a proof [6] that the minimum time between 2 ellipses grows as $1/\varepsilon$ for the planar 2-body problem. Here also, progress must be made. Explicit computation of the average system and its extremals for the 2-body problem has to be conducted.

Appendix. Proof of Proposition 3.16.

Proof of point 1. The integral in (37) is well defined (its integrand is bounded) and, by (36) and the Lebesgue convergence theorem, it is continuous with respect to X and h. Let us prove that this dH *is* the derivative of H. Since V is smooth, one has

$$\|\mathsf{V}(\theta,X+h) - \mathsf{V}(\theta,X) - \frac{\partial\mathsf{V}}{\partial X}(\theta,X).h\| \le k\,\|h\|^2\,, \tag{69}$$

where $\frac{\partial\mathsf{V}}{\partial X}(\theta,X)$ is smooth with respect to (θ,X) and k is some local constant. Now, assuming $\mathsf{V}(\theta,X) \neq 0$, one has

$$\begin{aligned}\|\mathsf{V}(\theta,X+h)\| - \|\mathsf{V}(\theta,X)\| = &\left(\mathsf{V}(\theta,X+h) - \mathsf{V}(\theta,X)\,\middle|\,\frac{\mathsf{V}(\theta,X)}{\|\mathsf{V}(\theta,X)\|}\right) \\ &+ a(\theta,X,h)\,\frac{\|\mathsf{V}(\theta,X+h)-\mathsf{V}(\theta,X)\|^2}{\|\mathsf{V}(\theta,X)\| + \|\mathsf{V}(\theta,X+h)\|}\end{aligned}$$

with $|a(\theta,X,h)| \le 2$. Hence, from (69) and (37), one has, for some local constant k',

$$\frac{\|\mathsf{H}(X+h) - \mathsf{H}(X) - \mathsf{dH}(X).h\|}{\|h\|} \le \frac{k'}{2\pi}\int_0^{2\pi}\left(\|h\| + \frac{\|\mathsf{V}(\theta,X+h)-\mathsf{V}(\theta,X)\|}{\|\mathsf{V}(\theta,X)\| + \|\mathsf{V}(\theta,X+h)\|}\right)\mathrm{d}\theta$$

for $\|h\|$ small enough. For fixed X and $h \to 0$, the integrand in the right-hand side is bounded by $1+\|h\|$ and converges to zero for θ outside the set $\{\theta \in S^1,\ \mathsf{V}(\theta,X) = 0\}$: by (36) and the Lebesgue convergence theorem, the right-hand side tends to zero. ∎

Let us now state two lemmas that are needed in the proof of point 2.

LEMMA A.1. *Assume that $\bar X \in \widetilde{\mathcal Z}$ and* (38) *is satisfied. There is a neighborhood U of $\bar X$ in $O^{\mathsf d}$ and a smooth map $\widehat\chi : U \to S^1$ such that, for $(\theta, X) \in U$, one has $\mathsf V(\theta,X) = 0$ only if $\theta = \widehat\chi(X)$, and*

$$\left(\frac{\partial \mathsf V}{\partial\theta}(\widehat\chi(X),X)\,\middle|\,\mathsf V(\widehat\chi(X),X)\right) = 0\,,\quad X \in U\,. \tag{70}$$

Proof. From (38.a), $\mathcal Z = \{(\theta,X) \in S^1 \times O^{\mathsf d}\,,\ \mathsf V(\theta,X) = 0\}$ is a smooth submanifold of $S^1 \times O^{\mathsf d}$, and from (38.c), $\widetilde{\mathcal Z}$ given by (35) is a smooth submanifold of $O^{\mathsf d}$, both of dimension $d+1-m$, and the projection $\pi : S^1 \times O^{\mathsf d} \to O^{\mathsf d}$ induces a diffeomorphism $\mathcal Z \to \widetilde{\mathcal Z}$ whose inverse is of the form $x \mapsto (\chi(x), x)$ with χ a smooth map $\widetilde{\mathcal Z} \to S^1$ that satisfies, for all $X \in \tilde Z$, $V(\theta,X) = 0$ if and only if $\theta = \chi(X)$.

Consider the map $T : S^1 \times O^{\mathsf d} \to \mathbb R$ given by $T(\theta,X) = \left(\frac{\partial \mathsf V}{\partial\theta}(\theta,X)\,\middle|\,\mathsf V(\theta,X)\right)$. Let $\bar X$ be in $\widetilde{\mathcal Z}$; since $\mathsf V(\chi(\bar X),\bar X) = 0$, one has $T(\chi(\bar X),\bar X) = 0$ and $\partial T/\partial\theta(\chi(\bar X),\bar X) = \|\frac{\partial \mathsf V}{\partial\theta}(\chi(\bar X),\bar X)\|^2$, nonzero from assumption (38.b): the implicit function theorem yields a unique map $\widehat\chi$ from a neighborhood U of $\bar X$ in $O^{\mathsf d}$ to a neighborhood of $\chi(\bar X)$ in S^1 such that $\theta = \widehat\chi(X)$ solves $T(\theta,X) = 0$; it must therefore coincide with χ in $U \cap \widetilde{\mathcal Z}$ and satisfies the lemma. □

LEMMA A.2. *Assume that $\bar X \in \widetilde{\mathcal Z}$ and* (38) *is satisfied. There exist a neighborhood U of $\bar X$ in $O^{\mathsf d}$, local coordinates $x_1,\ldots,x_d$ defined on U, and smooth maps $P\colon U \to SO(m)$, $\alpha\colon U \to \mathbb R$, and $W\colon S^1 \times U \to \mathbb R^m$ such that, with*

$$X_{\mathbf I} = \begin{pmatrix} x_1 \\ \vdots \\ x_{m-1} \end{pmatrix},$$

$$\mathsf V(\theta,X) = P(X)\left[\begin{pmatrix} X_{\mathbf I} \\ \alpha(X)\ (\theta - \widehat\chi(X)) \end{pmatrix} + (\theta - \widehat\chi(X))^2\ W(\theta,X)\right] \tag{71}$$

$$= P(X)\left[\begin{pmatrix} X_{\mathbf I} \\ 0 \end{pmatrix} + (\theta - \widehat\chi(X))\ W_1(\theta,X)\right] \tag{72}$$

$$\text{with}\ \ W_1(\theta,X) = \begin{pmatrix} 0_{m-1} \\ \alpha(X) \end{pmatrix} + (\theta - \widehat\chi(X))\ W(\theta,X) \tag{73}$$

in $S^1 \times U$, where α is bounded from below: $0 < \alpha_0 < \alpha(X)$, $X \in U$. Furthermore, for a constant $K_3 > 0$, one has, for all $(\theta,X) \in S^1 \times U$,

$$\|\mathsf V(\theta,X)\| \ge K_3\sqrt{\|X_{\mathbf I}\|^2 + \alpha(X)^2\,(\theta - \widehat\chi(X))^2}\,, \tag{74}$$

$$X_{\mathbf I} = 0\ \Rightarrow \|W_1(\theta,X)\| \ge K_3\,. \tag{75}$$

Proof. The map $X \mapsto \frac{\partial \mathsf V}{\partial\theta}(\widehat\chi(X),X)$ is nonzero for $X = \bar X$, and hence it does not vanish on a sufficiently small neighborhood U of $\bar X$, and one may write

$$\frac{\partial \mathsf V}{\partial\theta}(\widehat\chi(X),X)\ =\ P(X)\ \begin{pmatrix} 0_{m-1} \\ \alpha(X) \end{pmatrix},\quad \alpha(X) > \alpha_0 > 0\,. \tag{76}$$

Define $v_1,\ldots,v_m$, m smooth maps $S^1 \times U \to \mathbb R$ by

$$\begin{pmatrix} v_1(\theta,X) \\ \vdots \\ v_m(\theta,X) \end{pmatrix} = P^{-1}(X)\,\mathsf V(\theta,X)\,. \tag{77}$$

For i between 1 and $m-1$, $\frac{\partial v_i}{\partial\theta}(\widehat{\chi}(X),X)=0$ from (76), and $v_i(\widehat{\chi}(\bar{X}),\bar{X})=0$ from Lemma A.1, and, using (38.a), the rank of the map $X\mapsto(v_1(\widehat{\chi}(X),X),\ldots,v_{m-1}(\widehat{\chi}(X),X))$ is $m-1$ at $X=\bar{X}$: on a possibly smaller neighborhood U, there are local coordinates $x_1,\ldots,x_d$ such that $v_i(\theta,X)=x_i+(\theta-\widehat{\chi}(X))^2\,W_i(\theta,X)$ for $i\leq m-1$ and for some smooth W_i; substituting (76) and (77) in (70) implies $v_m(\widehat{\chi}(X),X)=0$, and hence $v_m(\theta,X)=\alpha(X)\,(\theta-\widehat{\chi}(X))+W_m(\theta,X)\,(\theta-\widehat{\chi}(X))^2$ for a smooth W_m; (71) is proved.

Possibly restricting U to a subset with compact closure, $\|W(\theta,X)\|$ is bounded on $S^1\times U$; if $|\theta-\widehat{\chi}(X)|\leq\frac{1}{2}\alpha_0/\max\|W\|$, then (74) holds with $K_3=\frac{1}{2}$ according to (71); on the set where $|\theta-\widehat{\chi}(X)|\geq\frac{1}{2}\alpha_0/\max\|W\|$, V does not vanish, and hence $(\|X_{\mathbf{I}}\|^2+\alpha(X)^2\,(\theta-\widehat{\chi}(X))^2)^{1/2}/\|\mathsf{V}(\theta,X)\|$ is bounded from below; (74) is proved, with K_3 smaller than both this bound and $\frac{1}{2}$. From (73), $W_1(\widehat{\chi}(\bar{X}),\bar{X})\neq 0$ because α does not vanish; from assumption (38.b) and (72) (where $X_{\mathbf{I}}=0$ if $X=\bar{X}$), $W_1(\theta,\bar{X})\neq 0$ if $\theta\neq\widehat{\chi}(\bar{X})$, and hence W_1 does not vanish on $S^1\times\{\bar{X}\}$; it is therefore bounded from below on $S^1\times U$ with U a small enough neighborhood of $\bar{X}$: (75) holds with K_3 smaller than this bound. $\square$

Proof of Proposition 3.16 *(point* 2*). We use* $[-\pi,\pi]$ *instead of* $[0,2\pi]$ *as an interval of integration.* Let $h\in\mathbb{R}^{\mathsf{d}}$, with $\|h\|=1$. From (37), one has, for some constant $\widetilde{K}$ using bounds on the derivatives of the smooth V,

$$\begin{aligned}|\mathrm{d}\mathsf{H}(X).h-\mathrm{d}\mathsf{H}(Y).h|\leq&\left|\frac{1}{2\pi}\int_{-\pi}^{\pi}\left(\frac{\partial\mathsf{V}}{\partial X}(\theta,X).h-\frac{\partial\mathsf{V}}{\partial X}(\theta,Y).h\left|\frac{\mathsf{V}(\theta,X)}{\|\mathsf{V}(\theta,X)\|}\right)\mathrm{d}\theta\right|\right.\\&+\left|\frac{1}{2\pi}\int_{-\pi}^{\pi}\left(\frac{\partial\mathsf{V}}{\partial X}(\theta,Y).h\left|\frac{\mathsf{V}(\theta,X)}{\|\mathsf{V}(\theta,X)\|}-\frac{\mathsf{V}(\theta,Y)}{\|\mathsf{V}(\theta,Y)\|}\right)\mathrm{d}\theta\right|\right.\\&\leq\widetilde{K}\|X-Y\|+\frac{\widetilde{K}}{2\pi}\left\|\int_{-\pi}^{\pi}\frac{\mathsf{V}(\theta,X)}{\|\mathsf{V}(\theta,X)\|}\mathrm{d}\theta-\int_{-\pi}^{\pi}\frac{\mathsf{V}(\theta,Y)}{\|\mathsf{V}(\theta,Y)\|}\mathrm{d}\theta\right\|.\end{aligned}$$

Finally, defining

$$\widehat{\mathsf{V}}(\varphi,X)=\mathsf{V}(\widehat{\chi}(X)+\varphi,X)\,,\quad\widehat{W_1}(\varphi,X)=W_1(\widehat{\chi}(X)+\varphi,X)\,,\tag{78}$$

and making a different change of variables in the last two integrals, one has

$$\begin{aligned}\|\mathrm{d}\mathsf{H}(X).h-\mathrm{d}\mathsf{H}(Y).h\|&\leq\widetilde{K}\|X-Y\|+\frac{\widetilde{K}}{2\pi}\int_{-\pi}^{\pi}\left\|\frac{\widehat{\mathsf{V}}(\varphi,X)}{\|\widehat{\mathsf{V}}(\varphi,X)\|}-\frac{\widehat{\mathsf{V}}(\varphi,Y)}{\|\widehat{\mathsf{V}}(\varphi,Y)\|}\right\|\mathrm{d}\varphi\\&\leq\widetilde{K}\|X-Y\|+\frac{\widetilde{K}}{\pi}\int_{-\pi}^{\pi}\frac{\|\widehat{\mathsf{V}}(\varphi,X)-\widehat{\mathsf{V}}(\varphi,Y)\|}{\|\widehat{\mathsf{V}}(\varphi,X)\|}\,\mathrm{d}\varphi,\end{aligned}\tag{79}$$

where the last inequality uses the fact $\|\frac{u}{\|u\|}-\frac{v}{\|v\|}\|\leq 2\min\{\frac{\|u-v\|}{\|u\|},\frac{\|u-v\|}{\|v\|}\}$; the inequality also holds with $\|\widehat{\mathsf{V}}(\varphi,Y)\|$ instead of $\|\widehat{\mathsf{V}}(\varphi,X)\|$ in the denominator. Now let us use Lemma A.2, and let $X=(x_1,\ldots,x_d)$ and $Y=(y_1,\ldots,y_d)$ in these coordinates; from (72), one has, with $\widehat{W_1}$ defined by (78),

$$\widehat{\mathsf{V}}(\varphi,X)=P(X)\left[\begin{pmatrix}X_{\mathbf{I}}\\0\end{pmatrix}+\varphi\,\widehat{W_1}(\varphi,X)\right],\ \widehat{\mathsf{V}}(\varphi,Y)=P(Y)\left[\begin{pmatrix}Y_{\mathbf{I}}\\0\end{pmatrix}+\varphi\,\widehat{W_1}(\varphi,Y)\right].\tag{80}$$

Hence

$$\begin{aligned}\widehat{\mathsf{V}}(\varphi,X)-\widehat{\mathsf{V}}(\varphi,Y)=&\big(P(X)-P(Y)\big)P(X)^{-1}\widehat{\mathsf{V}}(\varphi,X)\\&+P(Y)\left[\varphi\Big(W_1(\varphi,X)-W_1(\varphi,Y)\Big)+\begin{pmatrix}X_{\mathbf{I}}-Y_{\mathbf{I}}\\0\end{pmatrix}\right],\end{aligned}$$

and finally,

(81)
$$\frac{\|\widehat{\mathsf{V}}(\varphi,X)-\widehat{\mathsf{V}}(\varphi,Y)\|}{\|\widehat{\mathsf{V}}(\varphi,X)\|}\leq\|P(X)-P(Y)\|+\frac{|\varphi|\,\|W_1(\varphi,X)-W_1(\varphi,Y)\|}{\|\widehat{\mathsf{V}}(\varphi,X)\|}+\frac{\|X_{\mathbf{I}}-Y_{\mathbf{I}}\|}{\|\widehat{\mathsf{V}}(\varphi,X)\|}.$$

Two cases are to be distinguished:

(i) If $X_{\mathbf{I}}=Y_{\mathbf{I}}=0$, then φ factors out of $\widehat{\mathsf{V}}(\varphi,X)$ and $\widehat{\mathsf{V}}(\varphi,Y)$ in (80) and the last term in (81) is zero: according to (75), the integrand in (79) is bounded by

$$\|P(X)-P(Y)\|+\frac{\|\widehat{W}_1(\varphi,X)-\widehat{W}_1(\varphi,Y)\|}{K_3}\,,$$

and finally $|\mathsf{dH}(X).h-\mathsf{dH}(Y).h|\leq K\,\|X-Y\|$ with a constant K that depends only on V, the open set U, and the coordinates.

(ii) If $X_{\mathbf{I}}\neq 0$ (or $Y_{\mathbf{I}}\neq 0$, interchanging X and Y), then (81), using (74), implies that the integrand in (79) is bounded by

$$\|P(X)-P(Y)\|+\frac{1}{K_3}\,\frac{1}{\alpha_0}\;\|W_1(\varphi,X)-W_1(\varphi,Y)\|+\frac{1}{K_3}\sqrt{\frac{\|X_{\mathbf{I}}-Y_{\mathbf{I}}\|^2}{\|X_{\mathbf{I}}\|^2+\alpha(X)\varphi^2}}\,,$$

but the same is also true replacing $\alpha(X)$ with $\alpha(Y)$ and $\|X_{\mathbf{I}}\|^2$ with $\|Y_{\mathbf{I}}\|^2$; hence, since $\|a-b\|^2\leq 4\max\{\|a\|^2,\|b\|^2\}$, the last term may be replaced by $\frac{2}{K_3}\sqrt{\frac{\|X_{\mathbf{I}}-Y_{\mathbf{I}}\|^2}{\|X_{\mathbf{I}}-Y_{\mathbf{I}}\|^2+4\alpha_0\varphi^2}}$, whose integral between $-\pi$ and π is equal to

$$\frac{\|X_{\mathbf{I}}-Y_{\mathbf{I}}\|}{K_3\sqrt{\alpha_0}}\;\ln\left(1+\frac{4\pi\sqrt{\alpha_0}}{\|X_{\mathbf{I}}-Y_{\mathbf{I}}\|}+\frac{8\pi^2\alpha_0}{\|X_{\mathbf{I}}-Y_{\mathbf{I}}\|^2}\right),\tag{82}$$

which is less than $\|X_{\mathbf{I}}-Y_{\mathbf{I}}\|(k_1+k_2\ln\frac{1}{\|X_{\mathbf{I}}-Y_{\mathbf{I}}\|})$ for some k_1,k_2 when, say, $\frac{\|X_{\mathbf{I}}-Y_{\mathbf{I}}\|}{2\sqrt{\alpha_0}}<1$. Finally, since $\|X_{\mathbf{I}}-Y_{\mathbf{I}}\|$ is less than $\|X-Y\|$ and $u\mapsto u\ln(1/u)$ is nondecreasing, the expression (82) is bounded by $\|X-Y\|(k_1+k_2\,\ln\frac{1}{\|X-Y\|})$.

Cases (i) and (ii) imply (39), possibly restricting U so that $\ln\frac{1}{\|X-Y\|}\geq 1$. ∎

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